dynamic-networks · investigation report

# Dynamic Networks — a catalog of network-reconfiguring rules

running · viva-dynamic-networks · 15 studies · 2026-10-22

## Executive summary

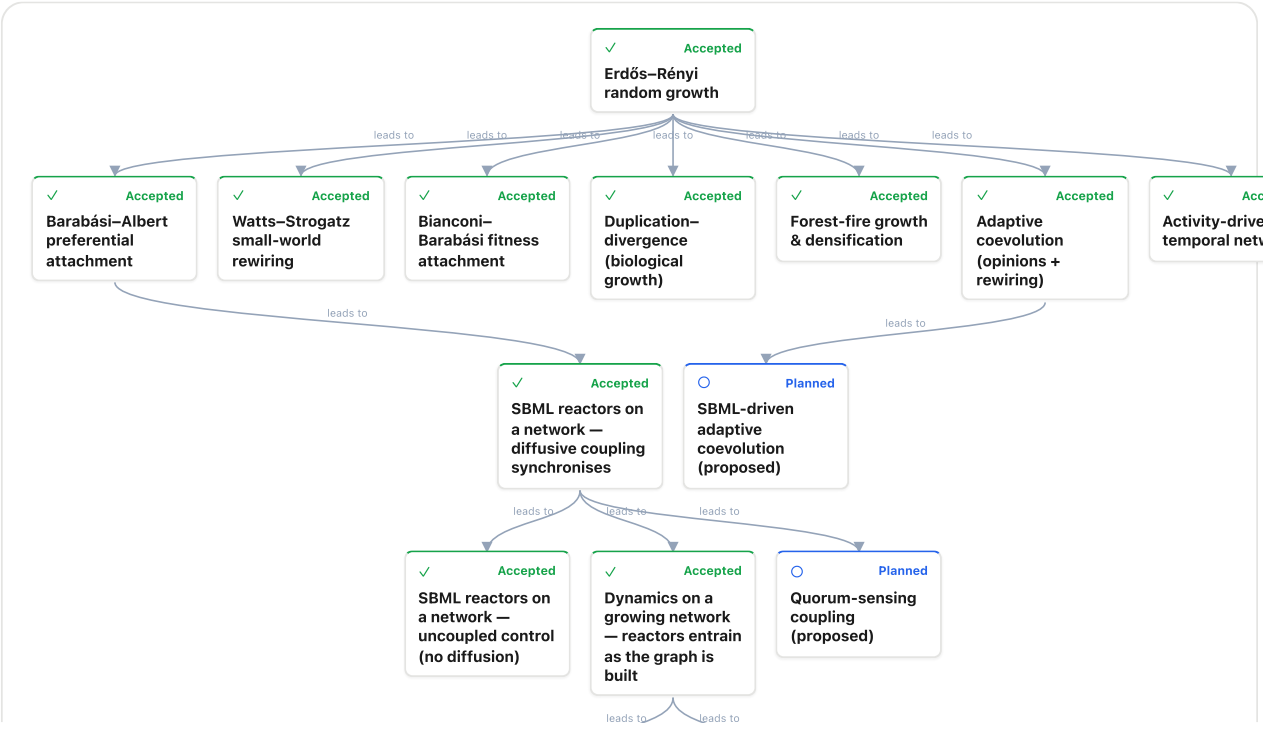
**Question.** How do different local rules reshape a network over time, and can a single process-bigraph process — driven by pluggable reactions — reproduce the canonical families of network dynamics, each with a clear visualization of the network evolving?

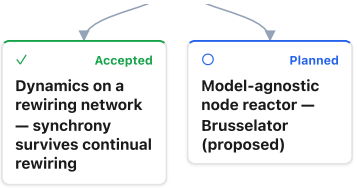
One generalized process reproduces all 8 Family-A dynamics; Family B runs a real SBML Repressilator (via viva-copasi) at every node with inter-node diffusion; and Family C runs that reaction–diffusion ON a topology the process is itself reconfiguring — growing or rewiring — so topology and node state co-evolve.

Across **15** studies — **12** supported · **0** qualified · **0** refuted · **3** planned. *Supported* = a computational gate passed under the stated model assumptions (evidence of feasibility, not proof the transition occurred).

## The investigation graph

The campaign as a dependency graph — each study a node badged with its code-computed status, wired to the studies it builds on.






**Status:** ✓ Accepted ◐ Investigating ○ Planned ⊘ Blocked ✕ Refuted

## Studies

Expand all | Collapse all

Every study in the campaign as a collapsible card — the question it tests, its verdict, and what it found at a glance. Click a card to expand the full record: model, runs, figures, tests, findings, and the limitations that bound it.

### ▾ 01 Erdős–Rényi random growth

● ACCEPTED

*What topology emerges when edges form purely at random, with no preference?*

**Hypothesis & discriminating control**

Random attachment yields a homogeneous Poisson degree distribution with no hubs — the null baseline for every structured rule that follows.

MODEL EXPERIMENT Run the erdos_renyi reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

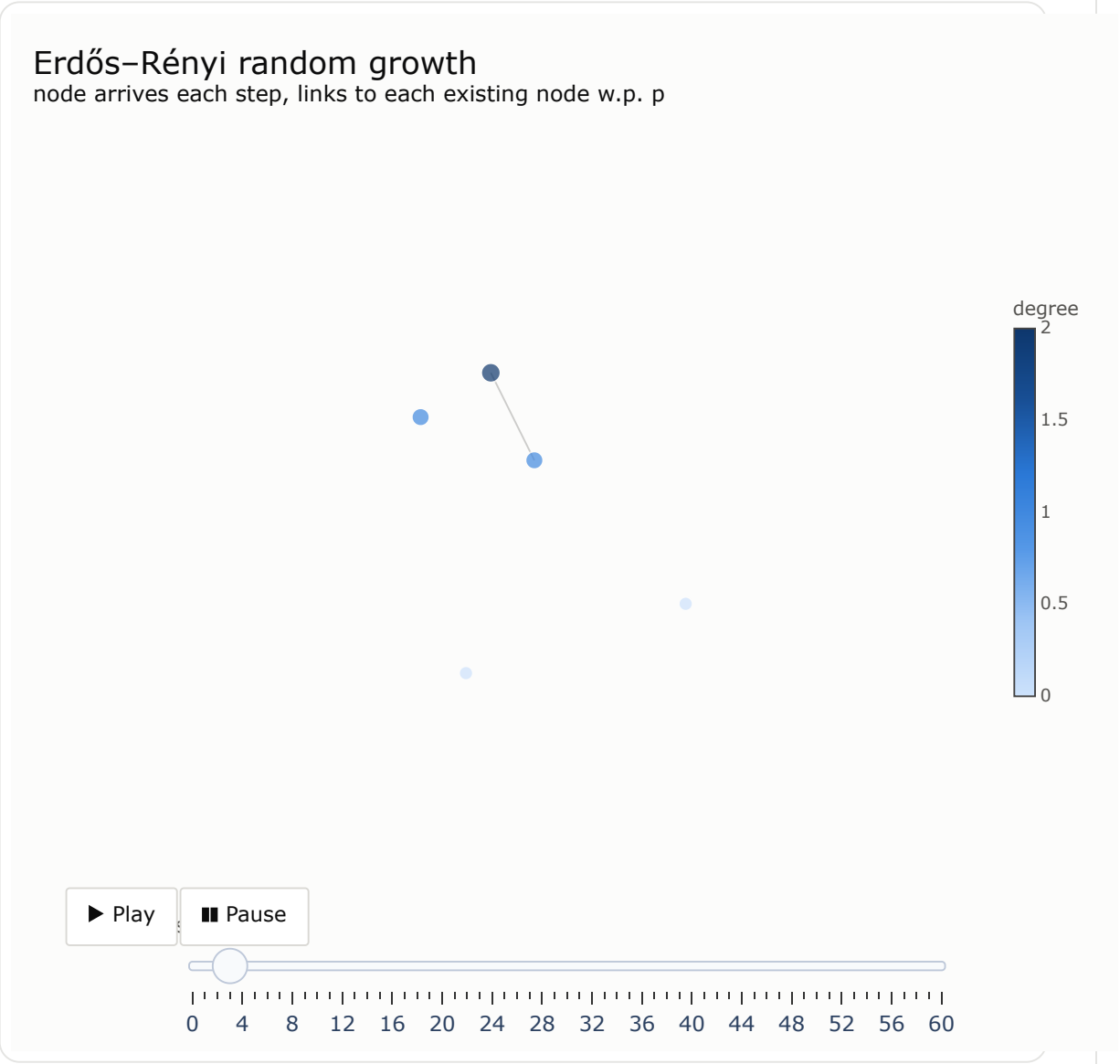
**Model & key assumptions**

**Methods · Simulations**

Model `erdos_renyi`

**Visualizations**

ERDOS RENYI




**Finding**

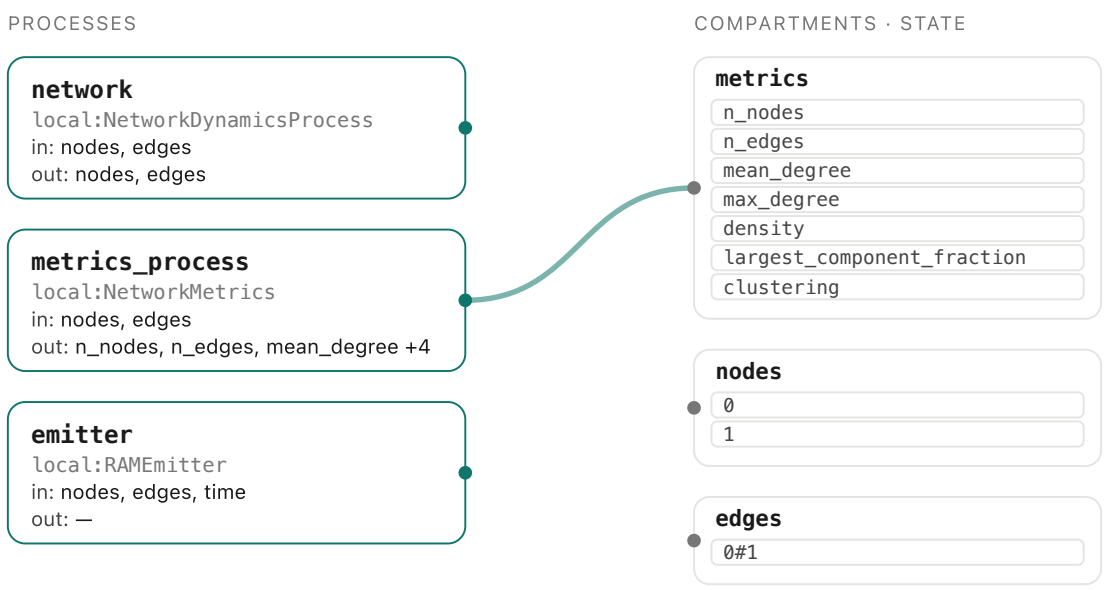
finding · Plausible

**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_a.erdos_renyi` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.erdos_renyi`

**Simulation run**

Reproduce: `vwb run study 01-erdos-renyi-random-growth`

**Tests**

| | | |
|---|---|---|
| A giant component spans ≥90% of nodes at the end.<br>`er-giant-component · quantitative · largest_component_fraction` | `at_least 0.9` | pending |
| Max degree is modest (< 4× mean degree) — no scale-free hub.<br>`er-no-dominant-hub · quantitative · max_degree` | `at_most 16` | pending |

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

## ▾ 02 Barabási–Albert preferential attachment ● ACCEPTED

*Does degree-proportional attachment produce scale-free hubs?*

**Hypothesis & discriminating control**

Linking new nodes preferentially to high-degree nodes yields a power-law degree distribution with a few dominant hubs.

MODEL EXPERIMENT Run the preferential_attachment reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

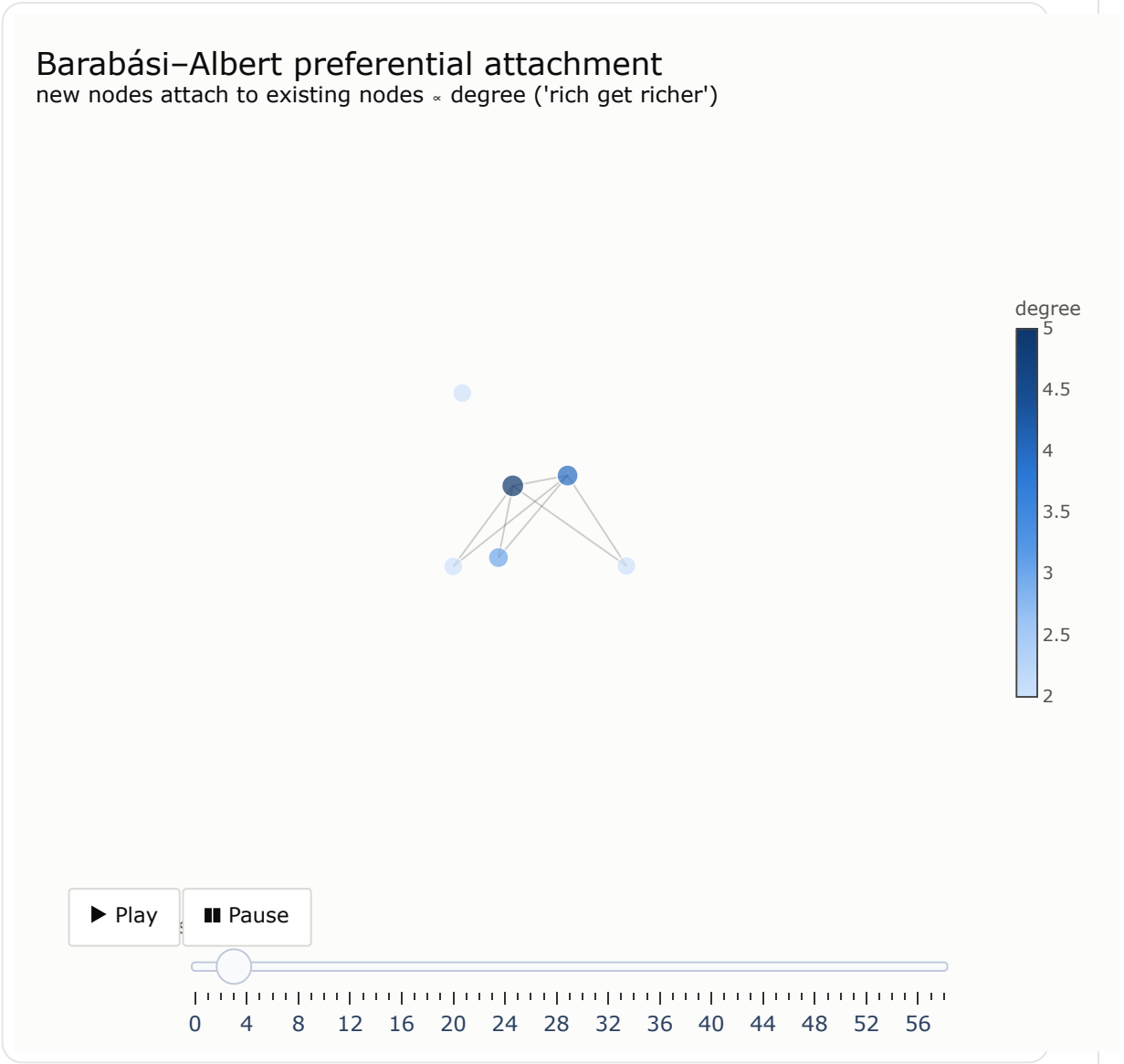
**Model & key assumptions**

**Methods · Simulations**

Model `preferential_attachment`

**Visualizations**

PREFERENTIAL ATTACHMENT




**Finding**

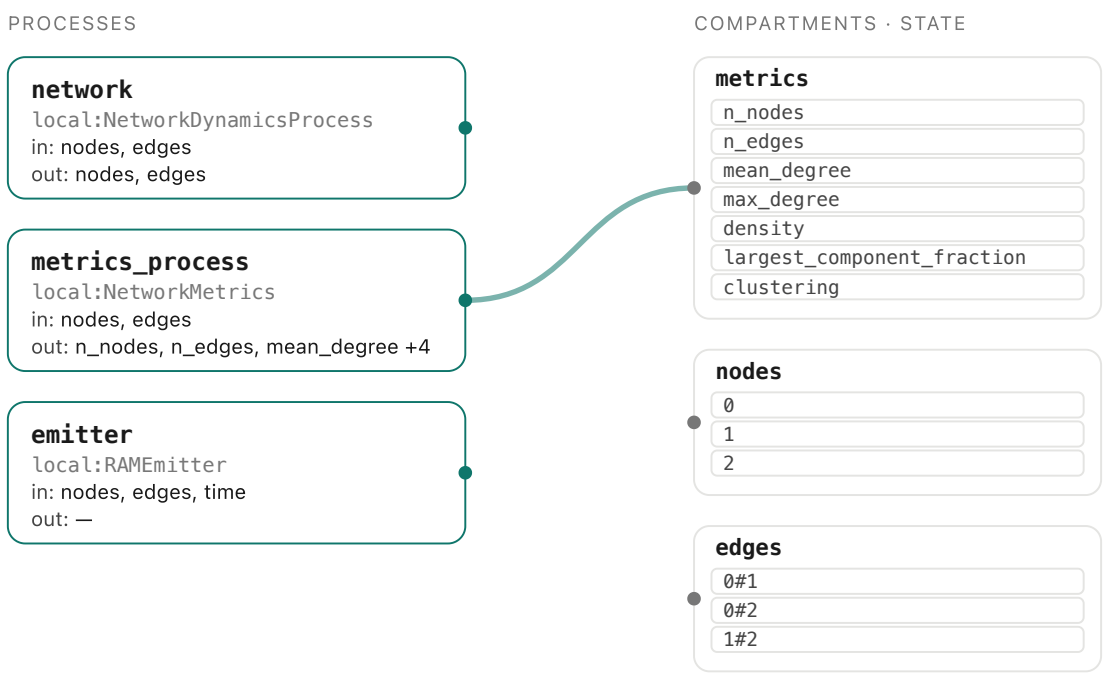
finding · Plausible

**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_a.preferential_attachment` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.preferential_attachment`

**Simulation run**

Reproduce: `vwb run study 02-preferential-attachment`

**Tests**

| | | |
|---|---|---|
| The maximum degree reaches ≥10 (a dominant hub).<br>`ba-hub-emerges · quantitative · max_degree` | `at_least 10` | pending |
| The network stays fully connected (BA growth never disconnects).<br>`ba-connected · quantitative · largest_component_fraction` | `at_least 0.99` | pending |

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

## ▾ 03 Watts–Strogatz small-world rewiring

● ACCEPTED

*How does random rewiring of a ring lattice change its structure?*

**Hypothesis & discriminating control**

Rewiring a small fraction of lattice edges collapses path length while clustering stays high — the small-world regime — without changing node or edge counts.

MODEL EXPERIMENT Run the small_world_rewire reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

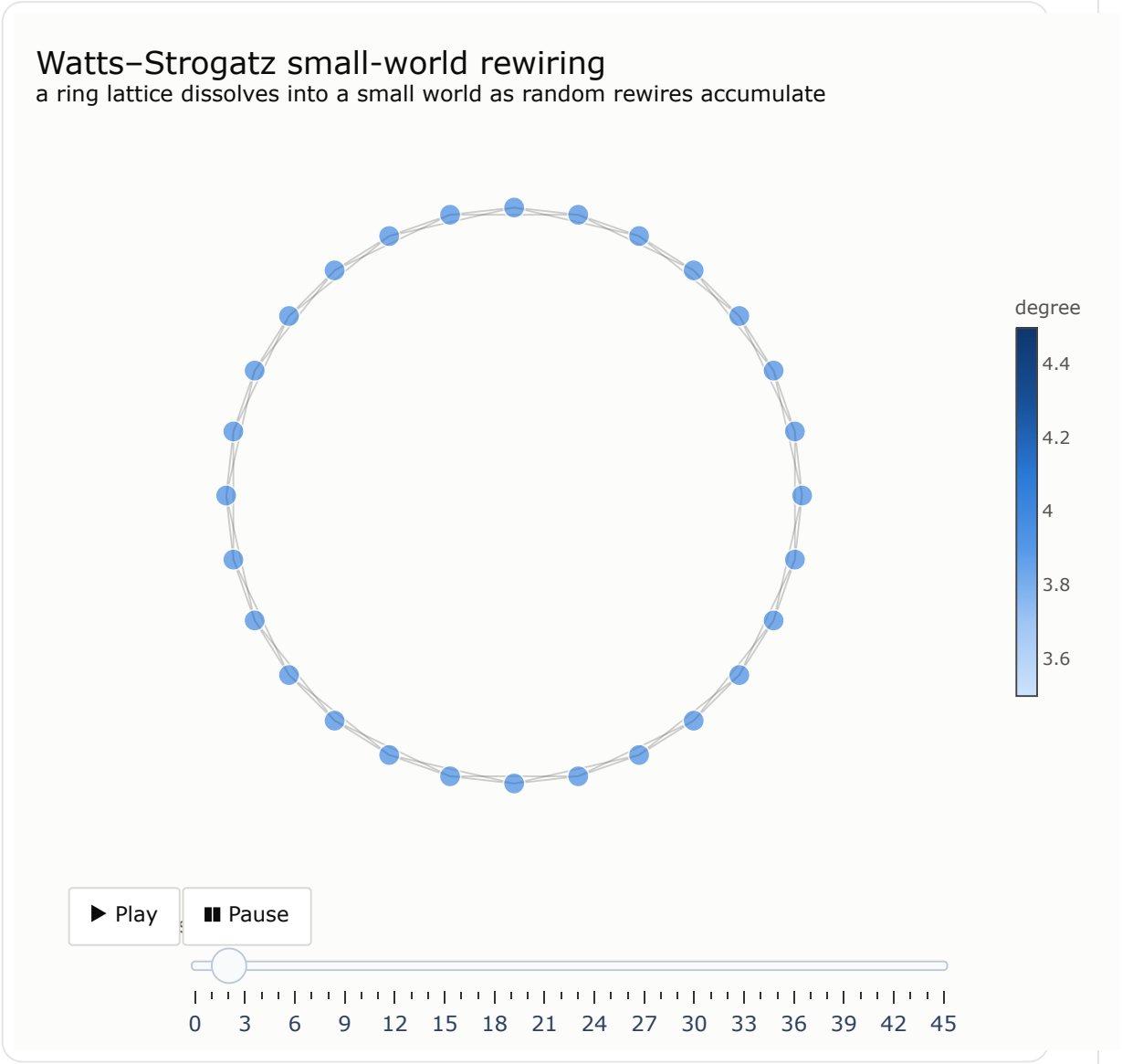
**Model & key assumptions**

**Methods · Simulations**

Model `small_world_rewire`

**Visualizations**

SMALL WORLD REWIRE




**Finding**

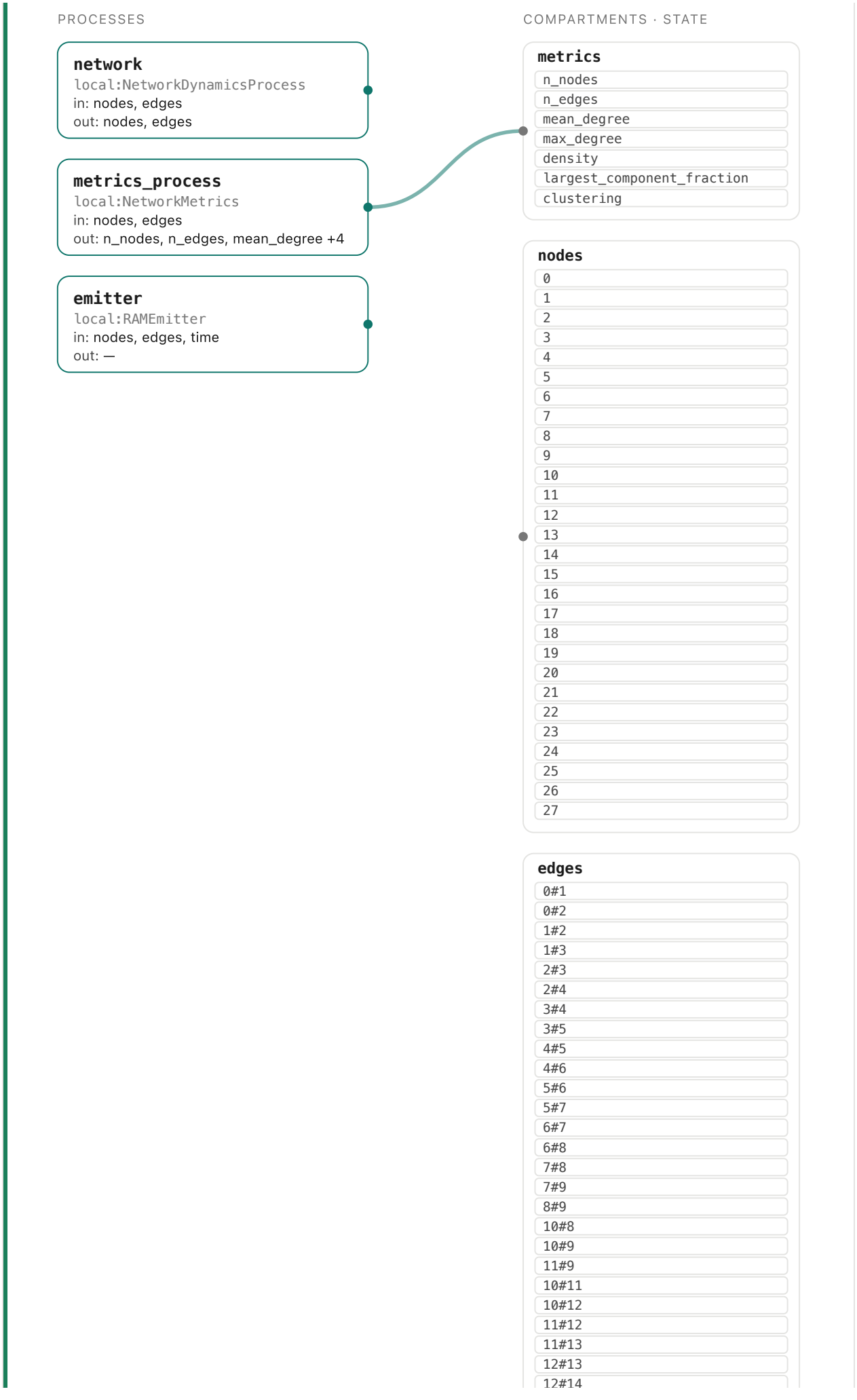
finding · Plausible

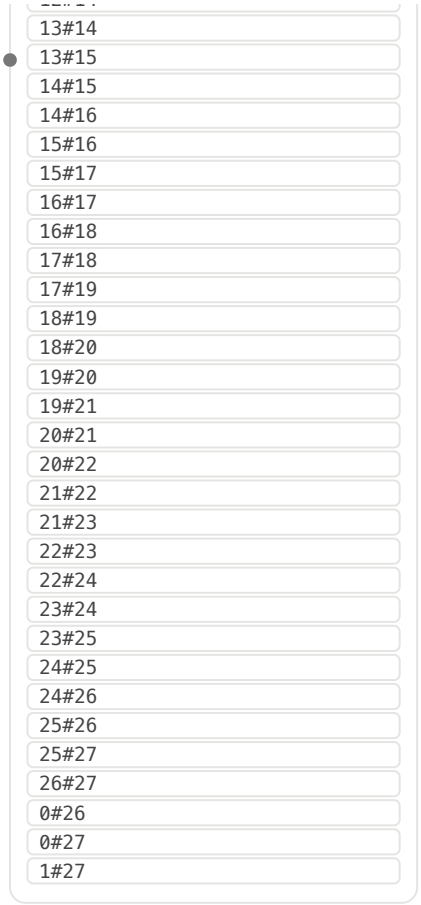
**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals


PROCESSES
network
local:NetworkDynamicsProcess
in: nodes, edges
out: nodes, edges
metrics_process
local:NetworkMetrics
in: nodes, edges
out: n_nodes, n_edges, mean_degree +4
emitter
local:RAMEmitter
in: nodes, edges, time
out: —
COMPARTMENTS · STATE
metrics
n_nodes
n_edges
mean_degree
max_degree
density
largest_component_fraction
clustering
nodes
0
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Static bigraph of `viva_dynamic_networks.composites.family_a.small_world_rewire` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.small_world_rewire`

**Simulation run**

Reproduce: `vwb run study 03-small-world-rewiring`

**Tests**

| | | |
|---|---|---|
| Edge count equals the initial ring-lattice edge count at the end.<br>`sw-edge-count-conserved · invariant · n_edges` | `equals 56` | pending |
| Node count is unchanged (no growth).<br>`sw-node-count-conserved · invariant · n_nodes` | `equals 28` | pending |

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

## ▾ 04 Bianconi–Barabási fitness attachment

● ACCEPTED

*Can an intrinsic node fitness let a latecomer overtake older, higher-degree nodes?*

**Hypothesis & discriminating control**

Weighting attachment by fitness×degree still produces hubs, but the leading hub need not be the oldest node — fitness breaks the pure first-mover advantage.

MODEL EXPERIMENT Run the fitness_attachment reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

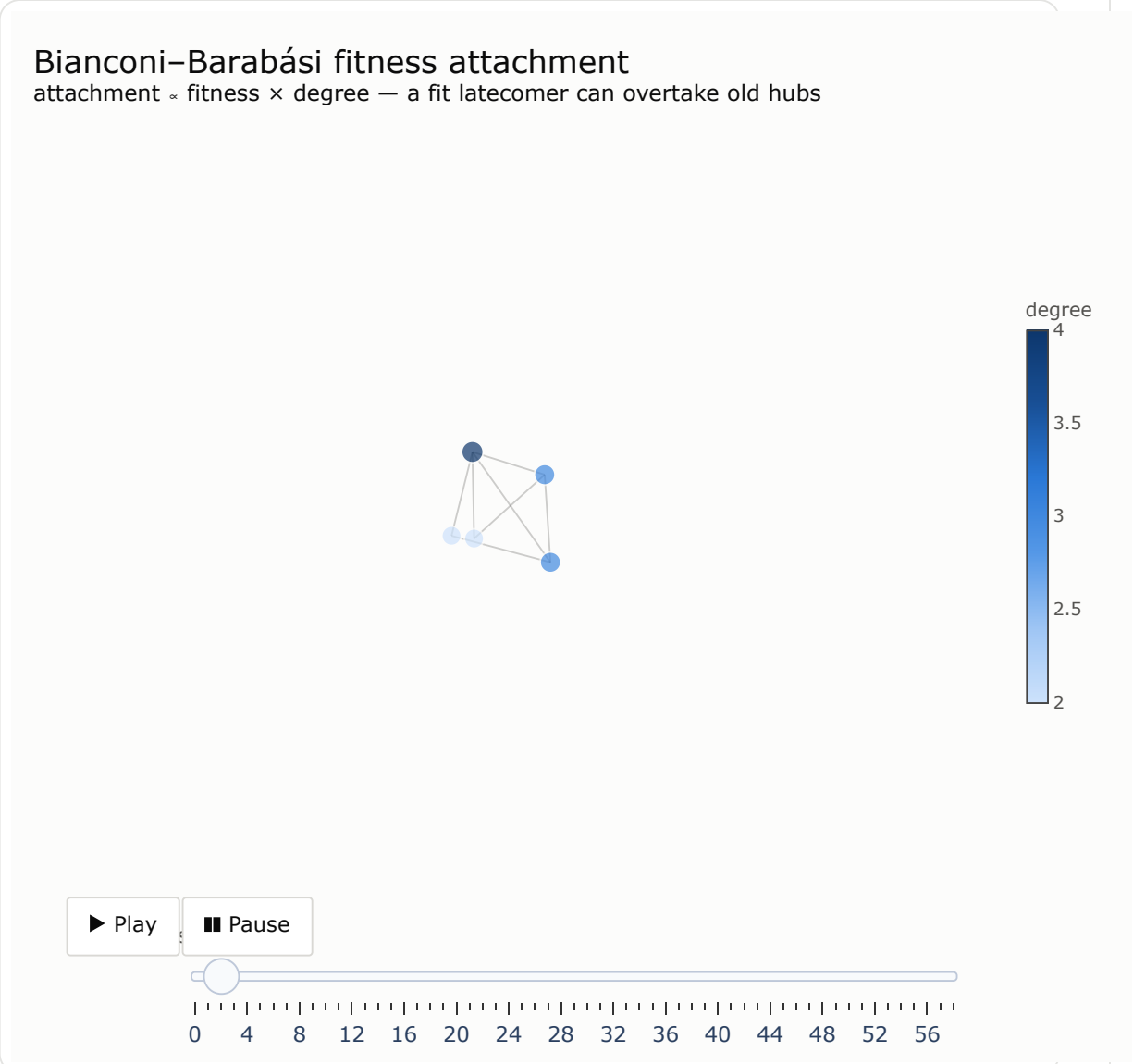
**Model & key assumptions**

**Methods · Simulations**

Model `fitness_attachment`

**Visualizations**

FITNESS ATTACHMENT




**Finding**

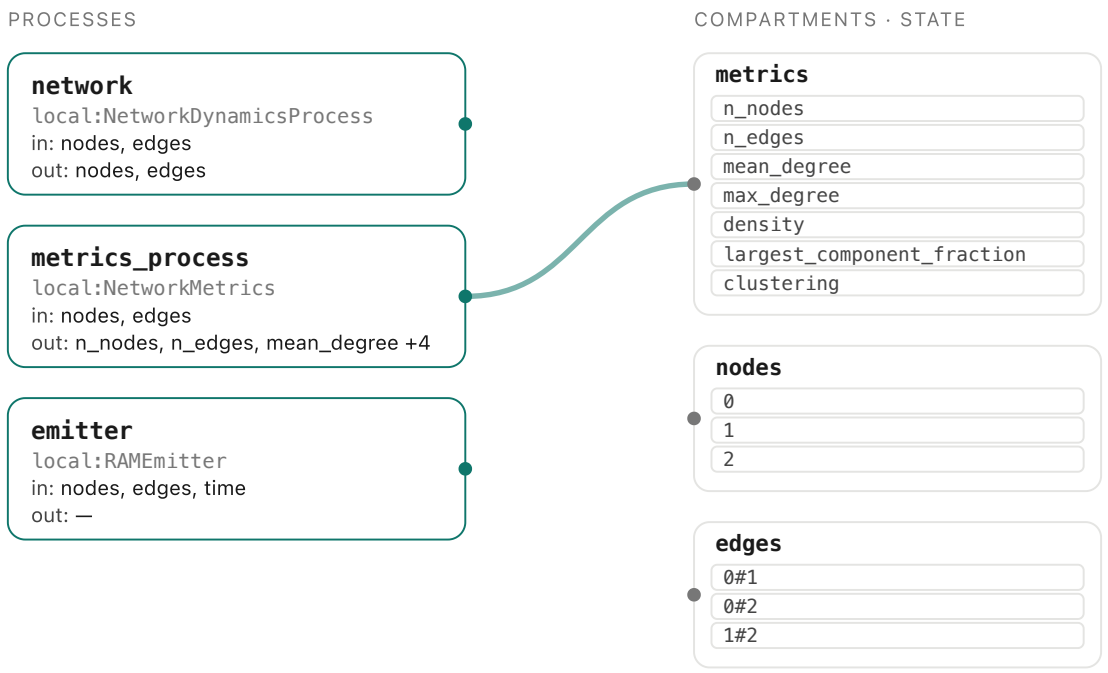
finding · Plausible

**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_a.fitness_attachment` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.fitness_attachment`

**Simulation run**

Reproduce: `vwb run study 04-fitness-condensation`

**Tests**

| | | |
|---|---|---|
| A dominant hub forms (max degree ≥10).<br>`fit-hub-emerges · quantitative · max_degree` | `at_least 10` | pending |
| The network stays connected.<br>`fit-connected · quantitative · largest_component_fraction` | `at_least 0.99` | pending |

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 05 # Duplication–divergence (biological growth)

● ACCEPTED

*Does genome-style duplication-and-divergence reproduce protein-network topology?*

**Hypothesis & discriminating control**

Copying a random node's neighbourhood and pruning edges yields a heavy-tailed, locally clustered network like observed protein-interaction graphs.

MODEL EXPERIMENT Run the duplication_divergence reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

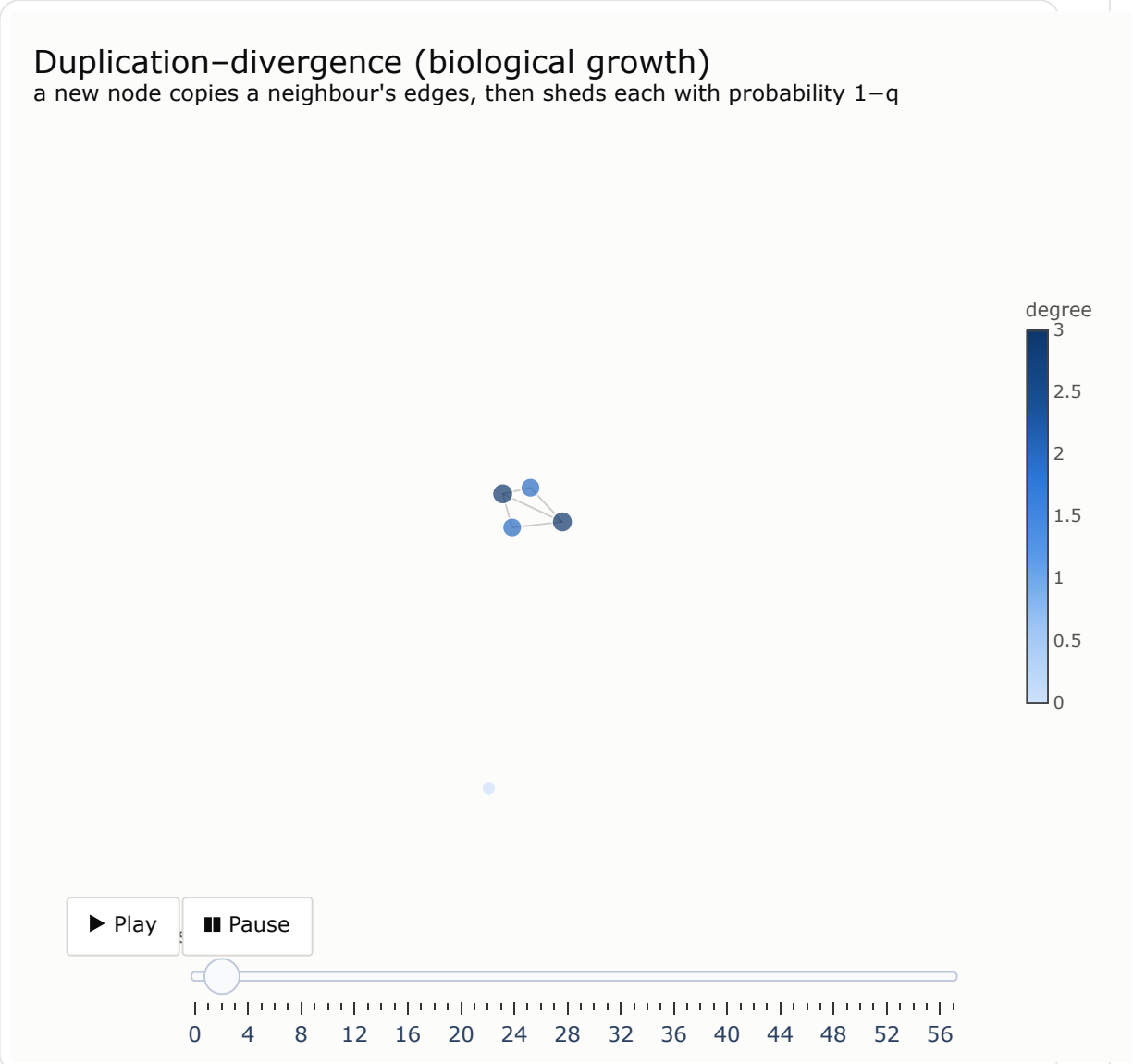
**Model & key assumptions**

**Methods · Simulations**

Model `duplication_divergence`

**Visualizations**

DUPLICATION DIVERGENCE




**Finding**

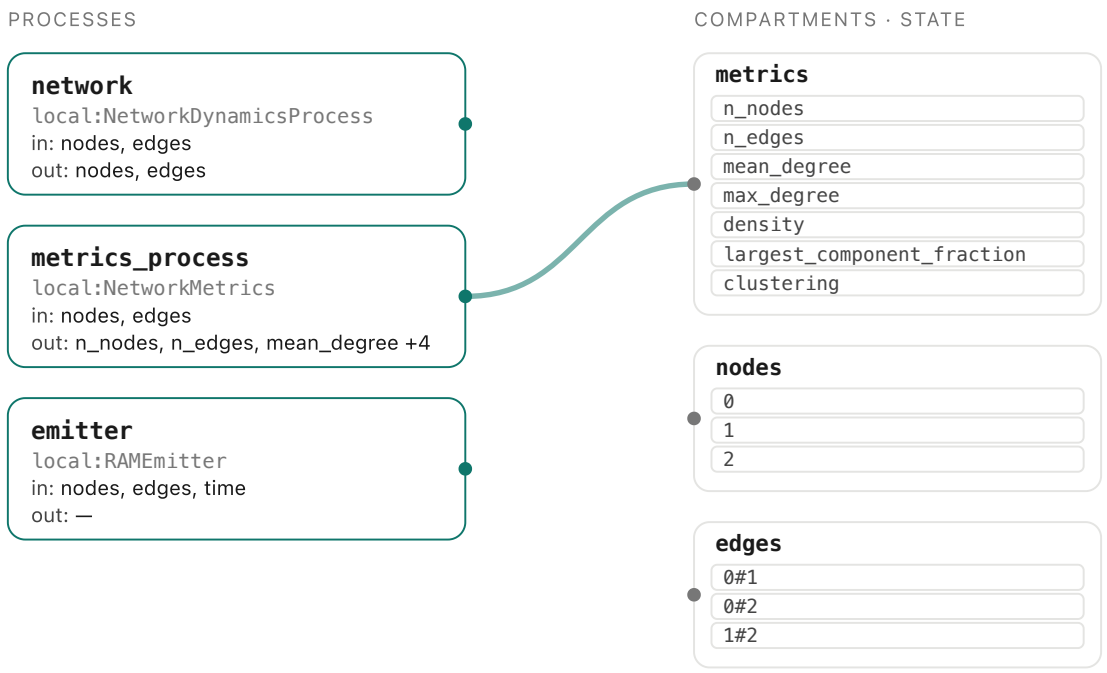
finding · Plausible

**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_a.duplication_divergence` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.duplication_divergence`

**Simulation run**

Reproduce: `vwb run study 05-duplication-divergence`

**Tests**

| | | |
|---|---|---|
| A heavy-tailed hub forms (max degree ≥6).<br>`dd-hub-forms · quantitative · max_degree` | `at_least 6` | pending |
| Average clustering is non-trivial (≥0.1).<br>`dd-clustered · quantitative · clustering` | `at_least 0.1` | pending |

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

## ▾ 06 Forest-fire growth & densification

● ACCEPTED

*Does recursive forest-fire burning densify the network over time?*

**Hypothesis & discriminating control**

Because each arrival can ignite many links, the edge count grows super-linearly in the node count — the network densifies and its mean degree rises.

MODEL EXPERIMENT Run the forest_fire reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

**Model & key assumptions**

**Methods · Simulations**

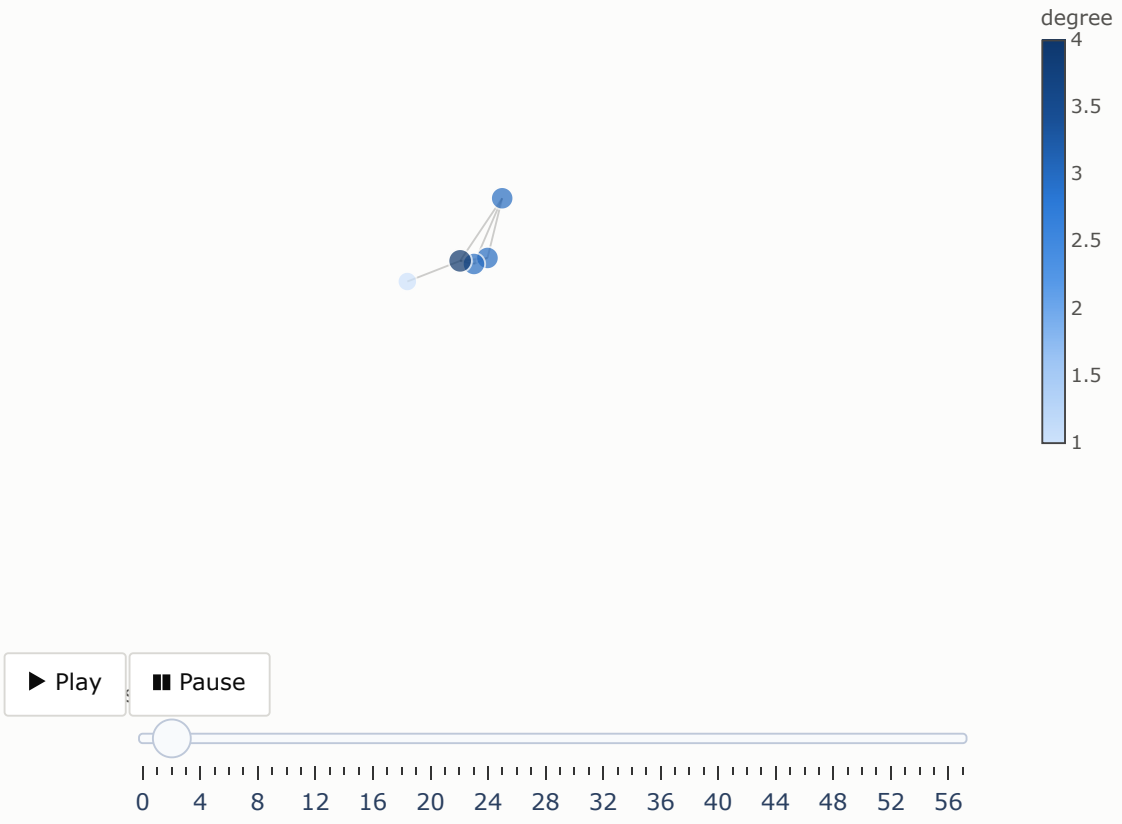
Model `forest_fire`

**Visualizations**

FOREST FIRE

## Forest-fire growth & densification

a new node 'burns' recursively outward, linking to a sampled subset of neighbours




**Finding**

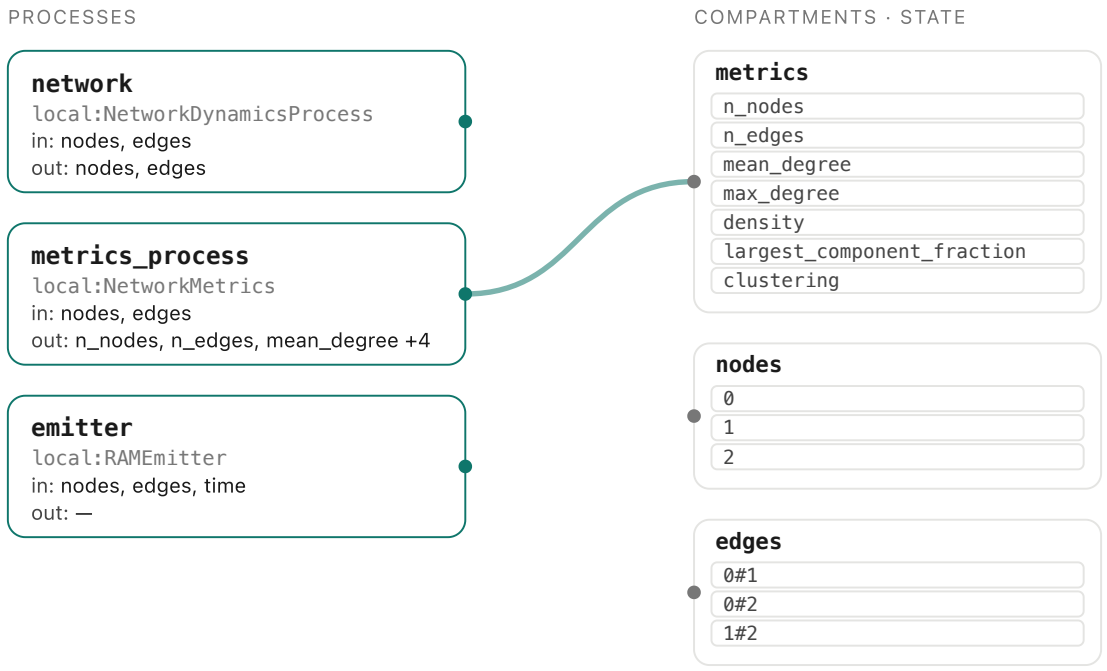
finding · Plausible

**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_a.forest_fire` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.forest_fire`

**Simulation run**

Reproduce: `vwb run study 06-forest-fire-densification`

**Tests**

| | | |
|---|---|---|
| Mean degree increases from start to end (densification).<br>`ff-densification · quantitative · mean_degree` | `at_least 3` | pending |
| The network stays connected.<br>`ff-connected · quantitative · largest_component_fraction` | `at_least 0.99` | pending |

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

## ▾ 07 Adaptive coevolution (opinions + rewiring)

● ACCEPTED

*When topology and node state co-adapt, does the network organise into like-minded communities?*

**Hypothesis & discriminating control**

Letting nodes rewire away from differing neighbours and toward similar ones (while opinions also spread) raises assortativity and segregates the two opinions — the coevolution bridge between Family A and dynamics-on-networks.

MODEL EXPERIMENT Run the adaptive_rewire reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

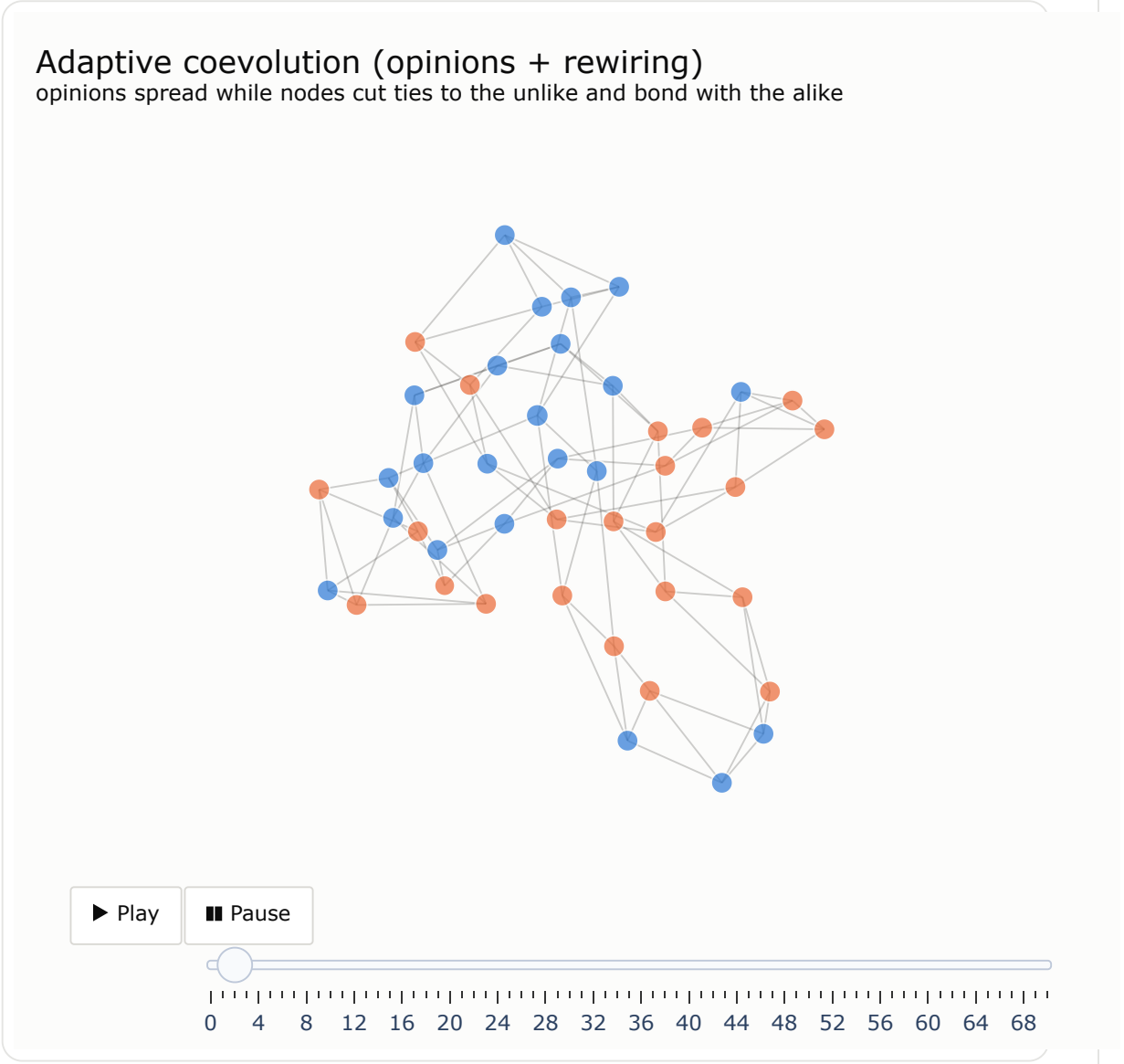
**Model & key assumptions**

**Methods · Simulations**

Model `adaptive_rewire`

**Visualizations**

ADAPTIVE REWIRE




**Finding**

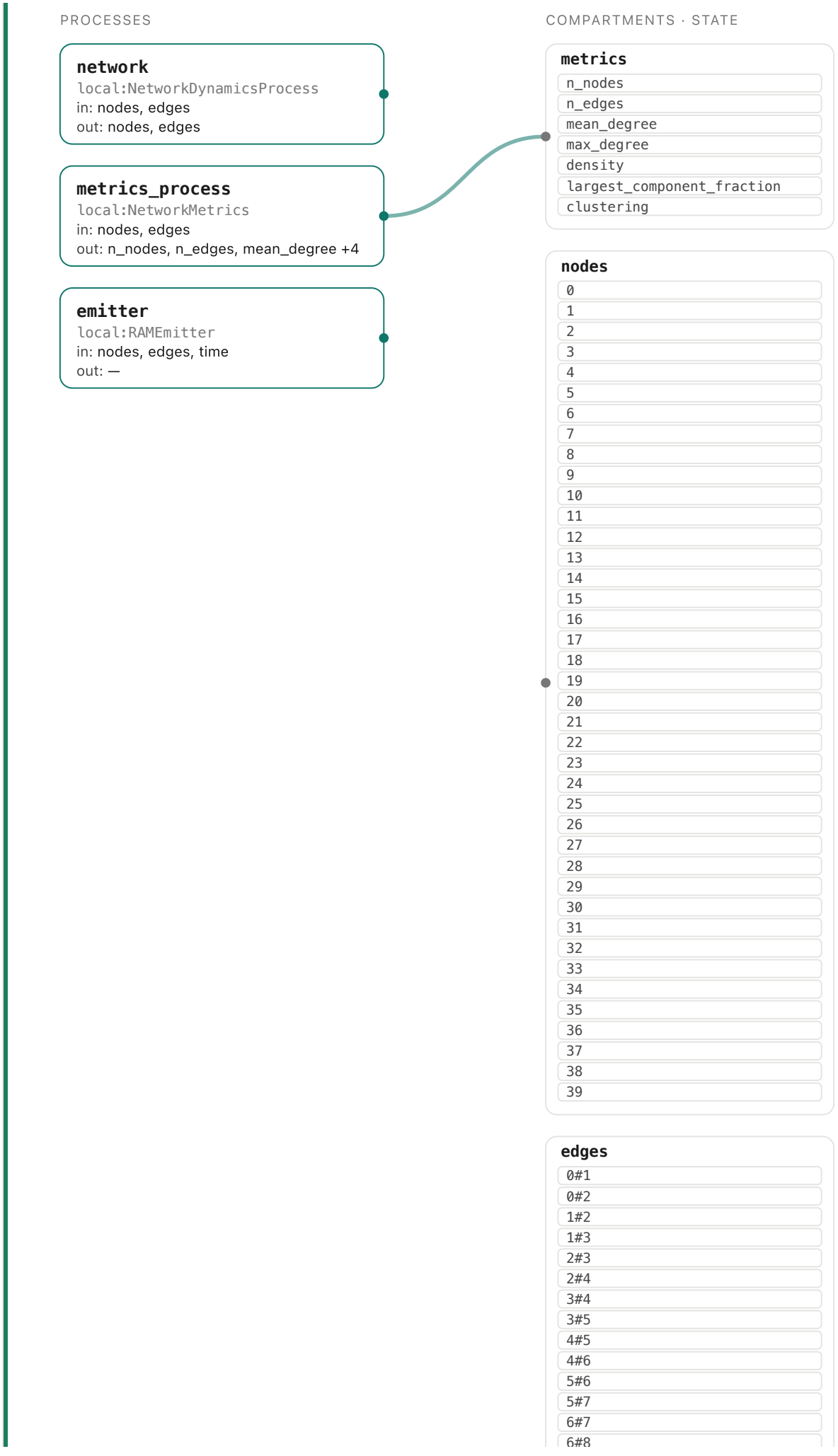
finding · Plausible

**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**


PROCESSES
network
local:NetworkDynamicsProcess
in: nodes, edges
out: nodes, edges
metrics_process
local:NetworkMetrics
in: nodes, edges
out: n_nodes, n_edges, mean_degree +4
emitter
local:RAMEmitter
in: nodes, edges, time
out: —
COMPARTMENTS · STATE
metrics
n_nodes
n_edges
mean_degree
max_degree
density
largest_component_fraction
clustering
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Static bigraph of `viva_dynamic_networks.composites.family_a.adaptive_rewire` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.adaptive_rewire`

**Simulation run**

Reproduce: `vwb run study 07-adaptive-coevolution`

**Tests**

| | | |
|---|---|---|
| Edge count is conserved (rewiring, not growth).<br>`ad-edge-count-conserved · invariant · n_edges` | `equals 88` | pending |
| Node count is conserved.<br>`ad-node-count-conserved · invariant · n_nodes` | `equals 44` | pending |

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

## ▾ 08 Activity-driven temporal network

● ACCEPTED

*What does an activity-driven contact sequence look like, where edges last only one time window?*

**Hypothesis & discriminating control**

With edges reset every step and nodes firing in proportion to their activity, each instantaneous graph is sparse and blinking, while the time-integrated degree follows the activity distribution.

MODEL EXPERIMENT Run the activity_driven reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

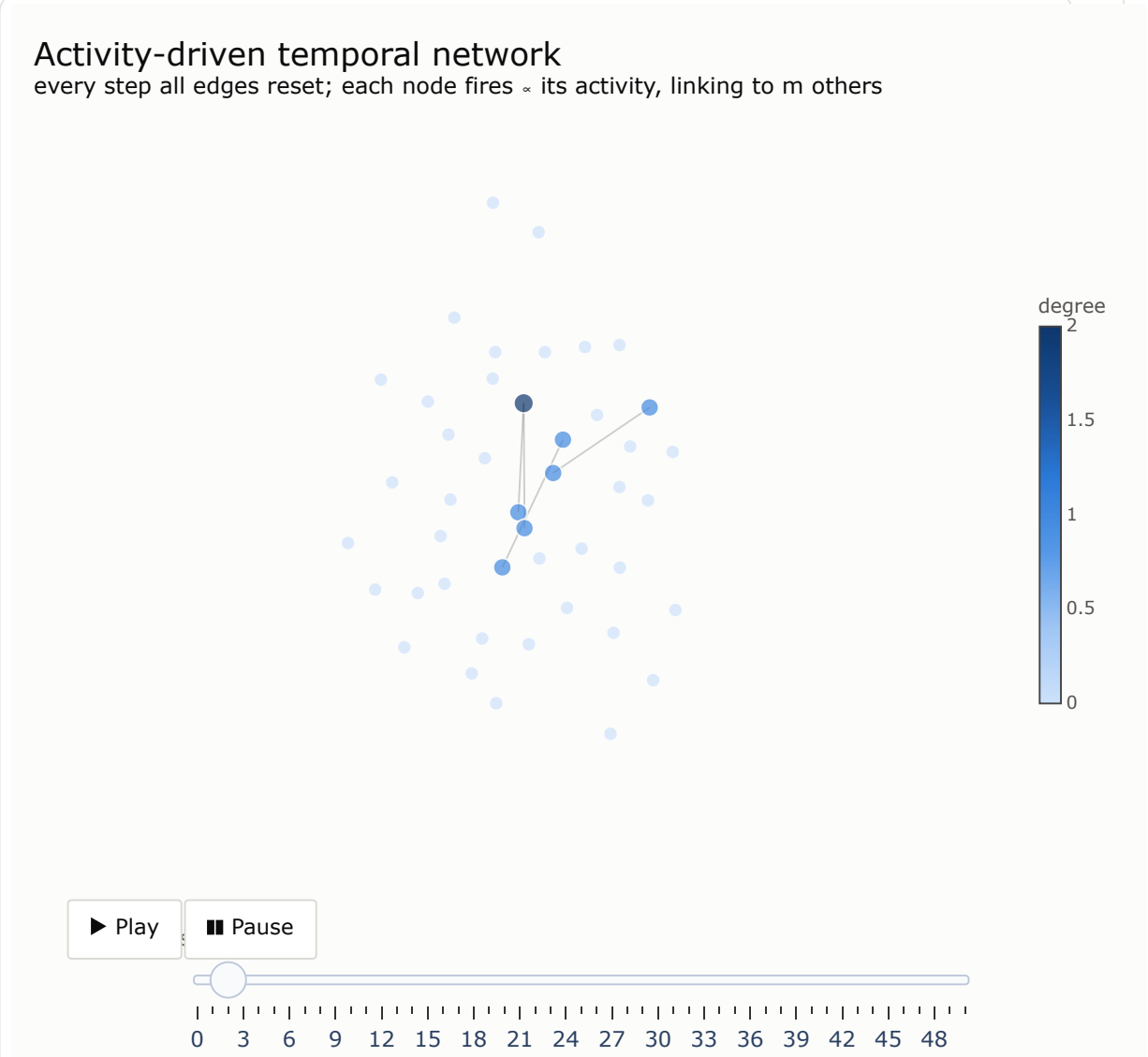
**Model & key assumptions**

**Methods · Simulations**

Model `activity_driven`

**Visualizations**

ACTIVITY DRIVEN




**Finding**

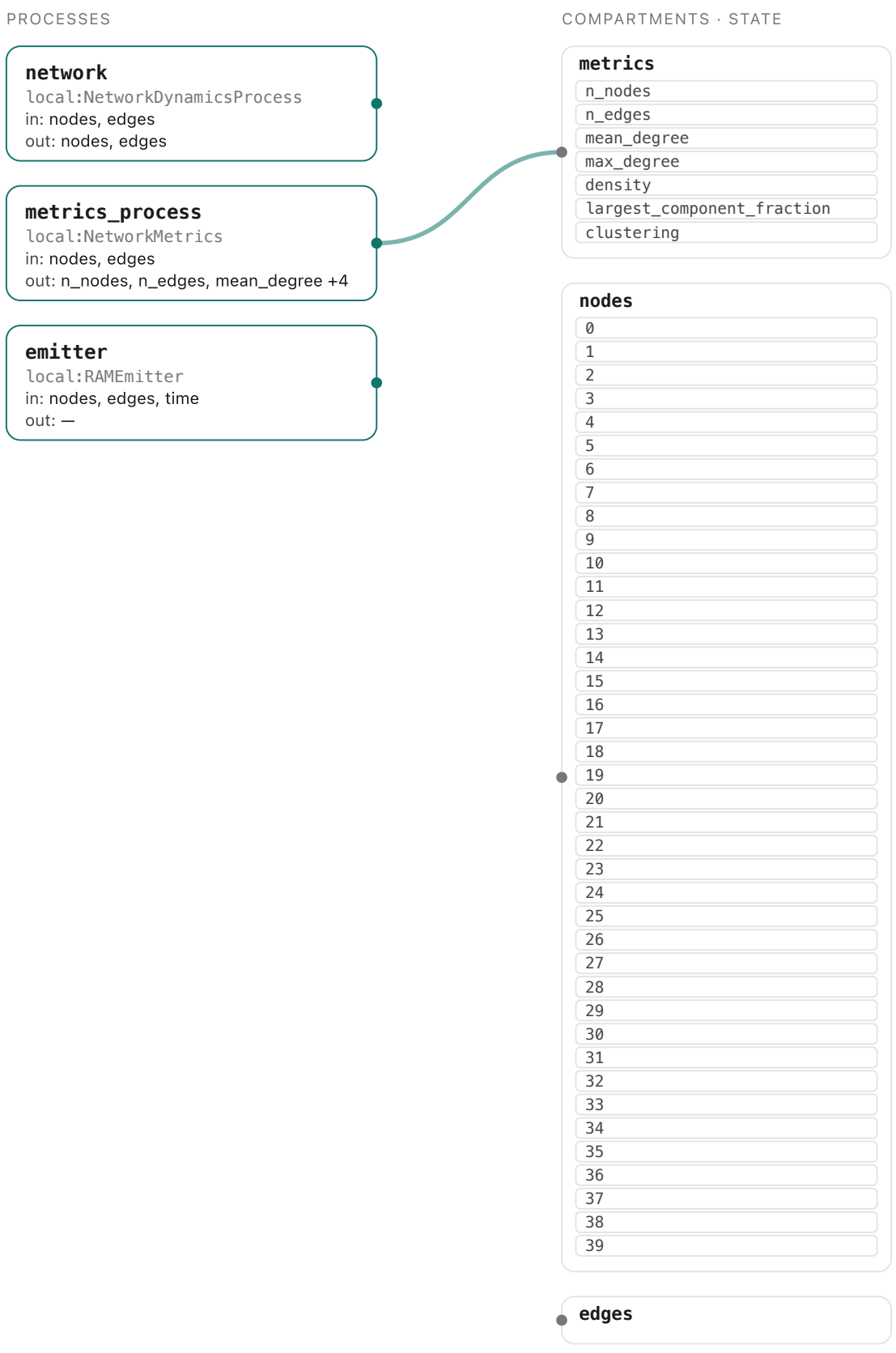
finding · Plausible

**Verdict**

● PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_a.activity_driven` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.activity_driven`

**Simulation run**

Reproduce: `vwb run study 08-activity-driven-temporal`

**Tests**

Node count is constant at 44.
`adn-node-count-conserved · invariant · n_nodes`

`equals 44` pending

Instantaneous snapshots stay sparse (mean degree < 2).
`adn-sparse-snapshot · quantitative · mean_degree`

`at_most 2` pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

## ▾ 09 SBML reactors on a network — diffusive coupling synchronises

ACCEPTED

*When a continuous SBML oscillator runs at every node and species diffuse along the edges, does the network synchronise?*

**Hypothesis & discriminating control**

Diffusive coupling of per-node Repressilators pulls their phases together, so the across-node spread of a protein collapses toward zero — synchronisation.

MODEL EXPERIMENT Run the repressilator_network_coupled composite (SBML reactors + inter-node transport) and measure across-node synchrony.

OBSERVABLE `1 acceptance test`

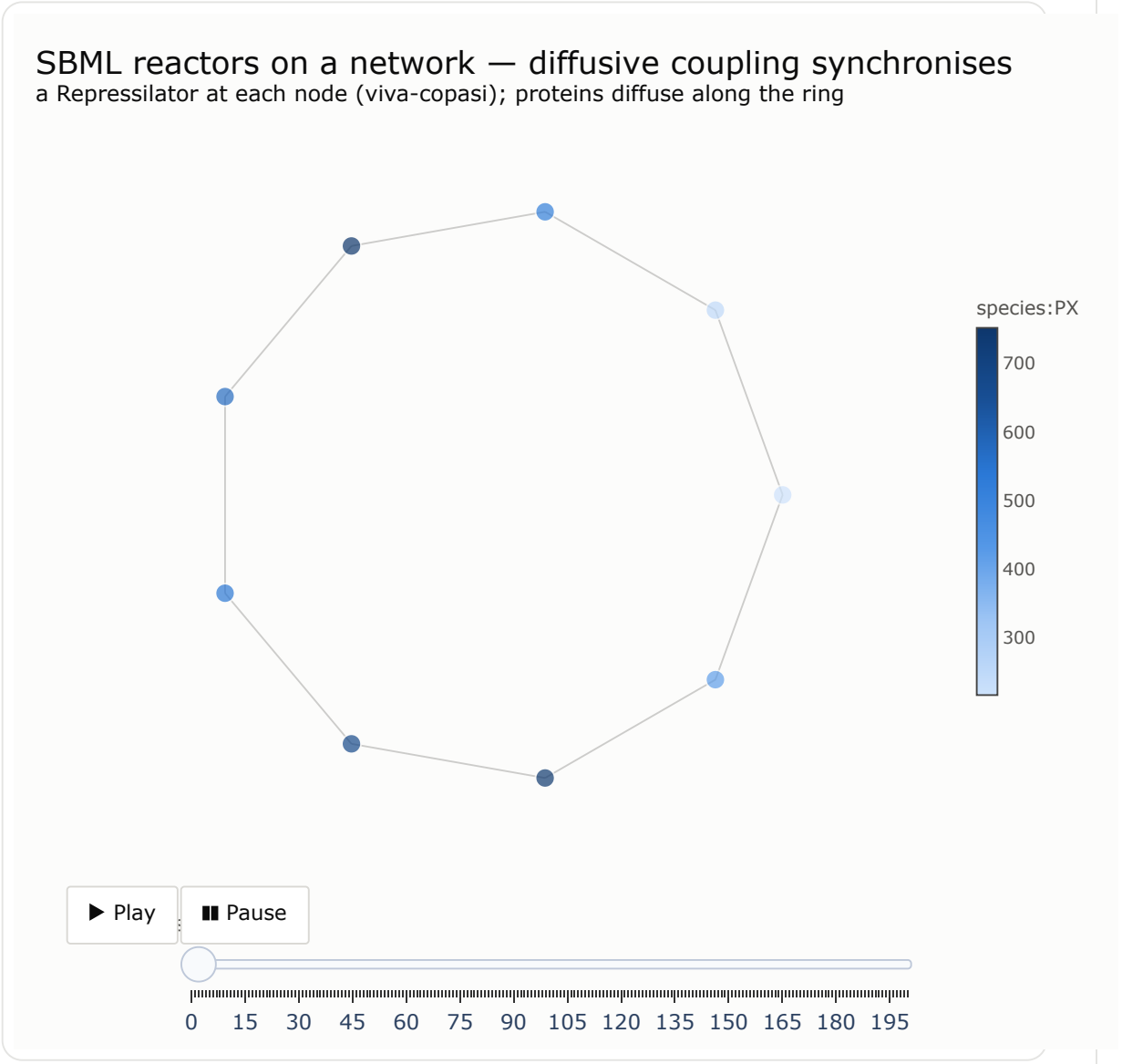
**Model & key assumptions**

**Methods · Simulations**

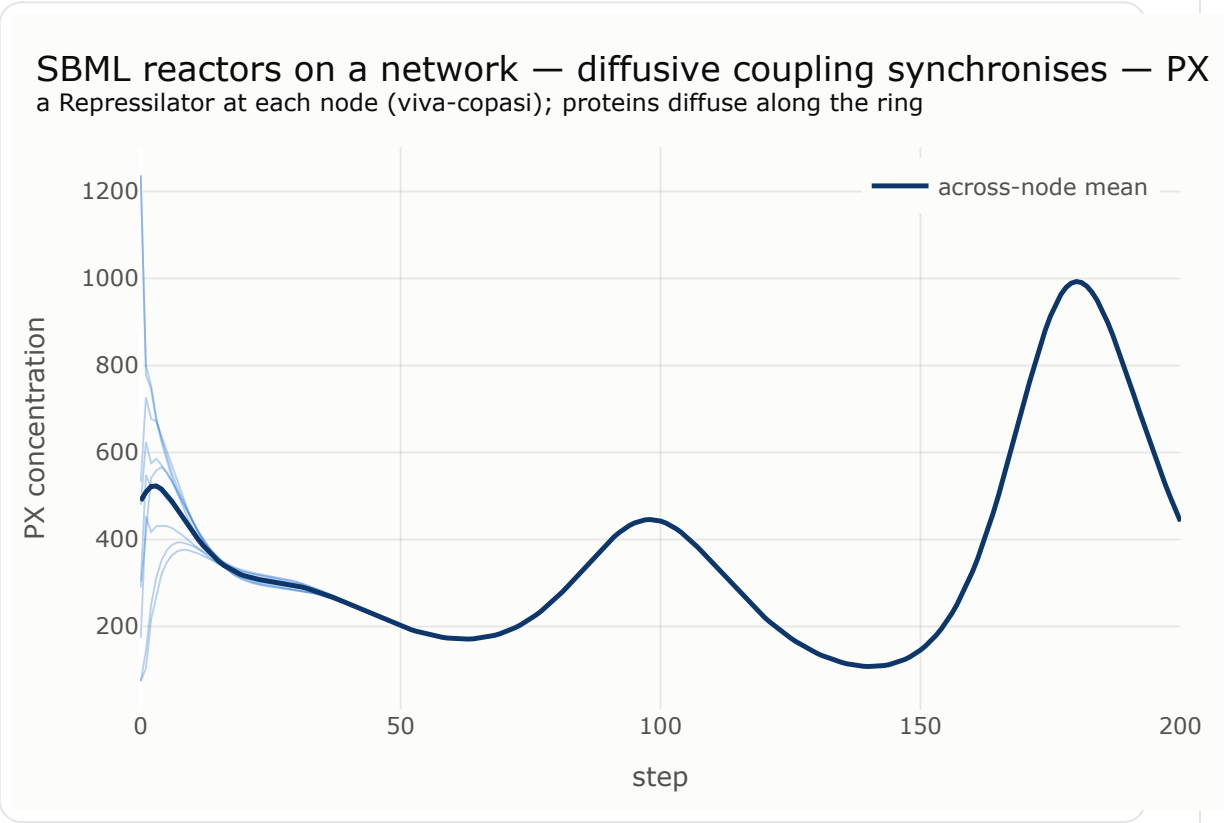
Model `repressilator_network_coupled`

**Visualizations**

REPRESSILATOR NETWORK COUPLED



REPRESSILATOR NETWORK COUPLED TIMESERIES




**Finding**

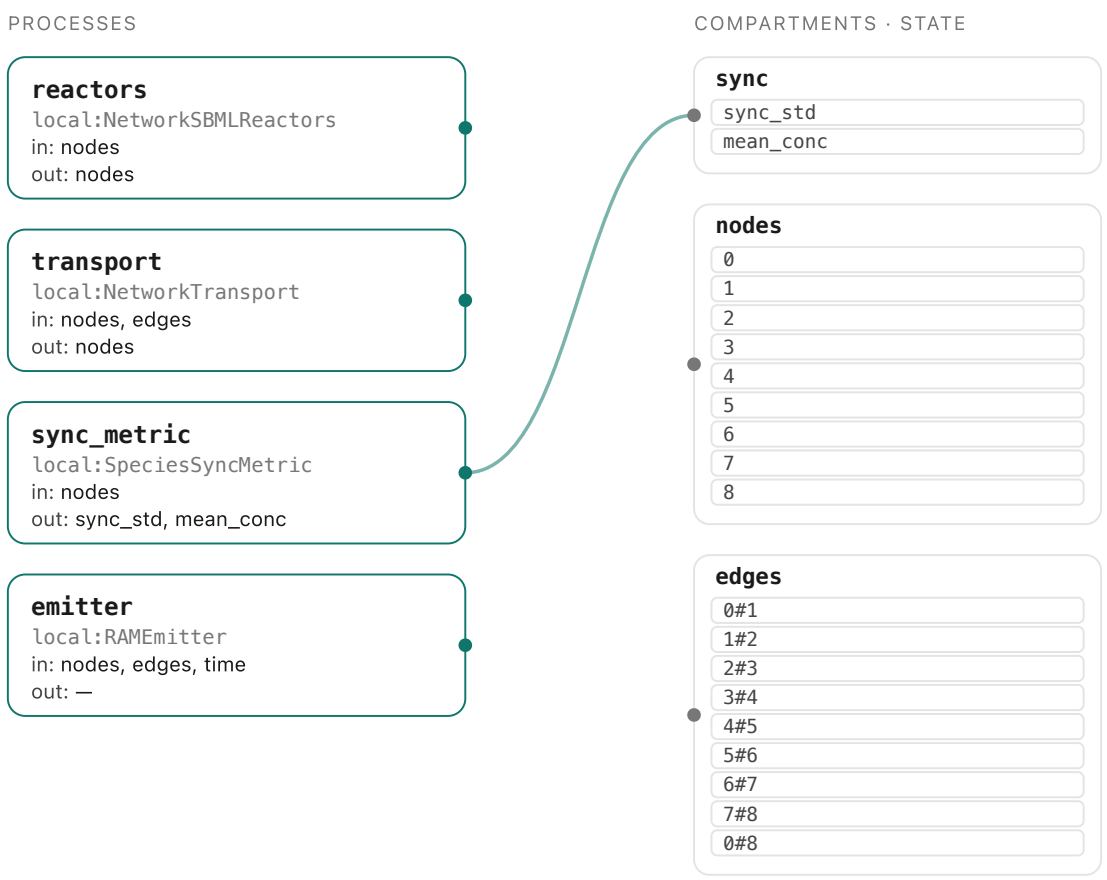
finding · Plausible

**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_b.repressilator_network_coupled` — 4 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_b.repressilator_network_coupled`

**Simulation run**

Reproduce: `vwb run study 09-sbml-nodes-coupled`

**Tests**

Final across-node PX spread is <25% of its initial value.
`coupled-synchronises · quantitative · sync_std`

`at_most 0.25` pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

▾ 10

## SBML reactors on a network — uncoupled control (no diffusion)

● ACCEPTED

*With diffusion switched off, do the per-node oscillators stay out of phase?*

**Hypothesis & discriminating control**

Without inter-node transport each Repressilator runs independently, so nodes seeded at different phases never synchronise and the across-node spread stays

large.

MODEL EXPERIMENT Run the repressilator_network_uncoupled composite (SBML reactors + inter-node transport) and measure across-node synchrony.

OBSERVABLE `1 acceptance test`

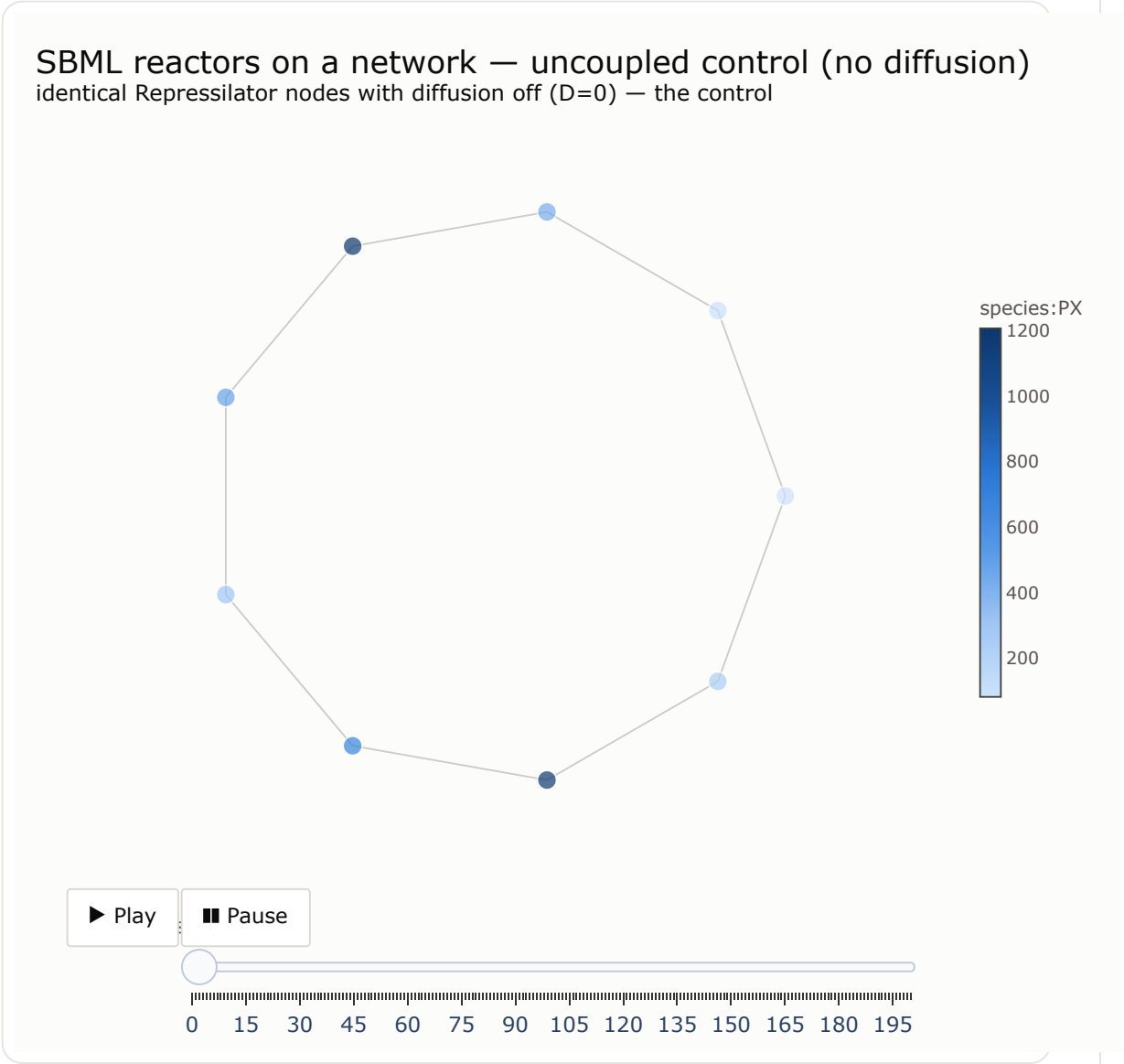
**Model & key assumptions**

**Methods · Simulations**

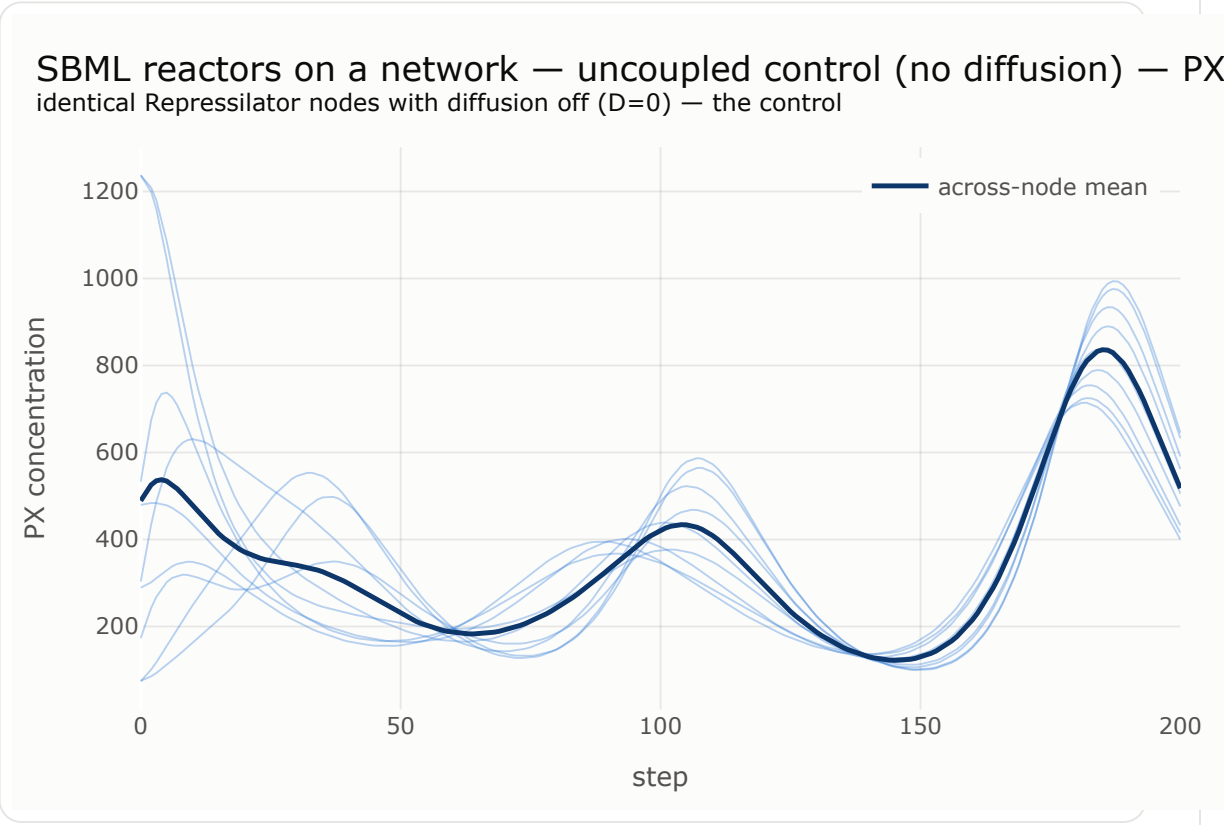
Model `repressilator_network_uncoupled`

**Visualizations**

REPRESSILATOR NETWORK UNCOUPLED



REPRESSILATOR NETWORK UNCOUPLED TIMESERIES




**Finding**

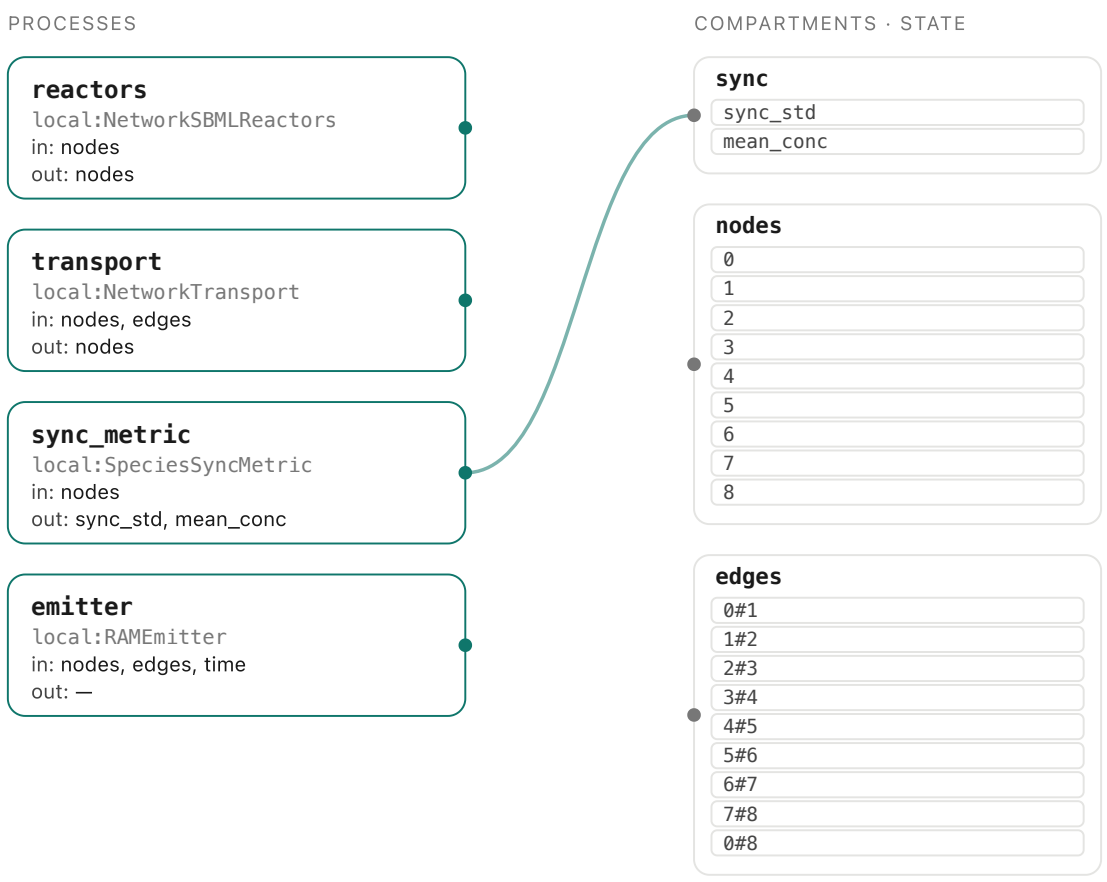
finding · Plausible

**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_b.repressilator_network_uncoupled` — 4 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_b.repressilator_network_uncoupled`

**Simulation run**

Reproduce: `vwb run study 10-sbml-nodes-uncoupled`

**Tests**

Final across-node PX spread stays well above the coupled case (ratio not collapsed). `at_least 3` pending

`uncoupled-stays-spread · quantitative · sync_std`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

## ▾ 11 Dynamics on a growing network — reactors entrain as the graph is built

● ACCEPTED

*Can SBML reactor dynamics run on a network whose topology is itself being built by a process-bigraph reaction, and do newcomers entrain?*

**Hypothesis & discriminating control**

Running the Family-B reactors + transport on a graph that a Family-A growth reaction is simultaneously building, each newcomer joins at its own phase and is pulled into synchrony by the dense local coupling — the full synthesis of dynamics OF and ON the network.

MODEL EXPERIMENT Run the growing_repressilator_network composite — SBML reactors + transport on a topology that a PBG reaction reconfigures during the run — and measure synchrony.

OBSERVABLE `1 acceptance test`

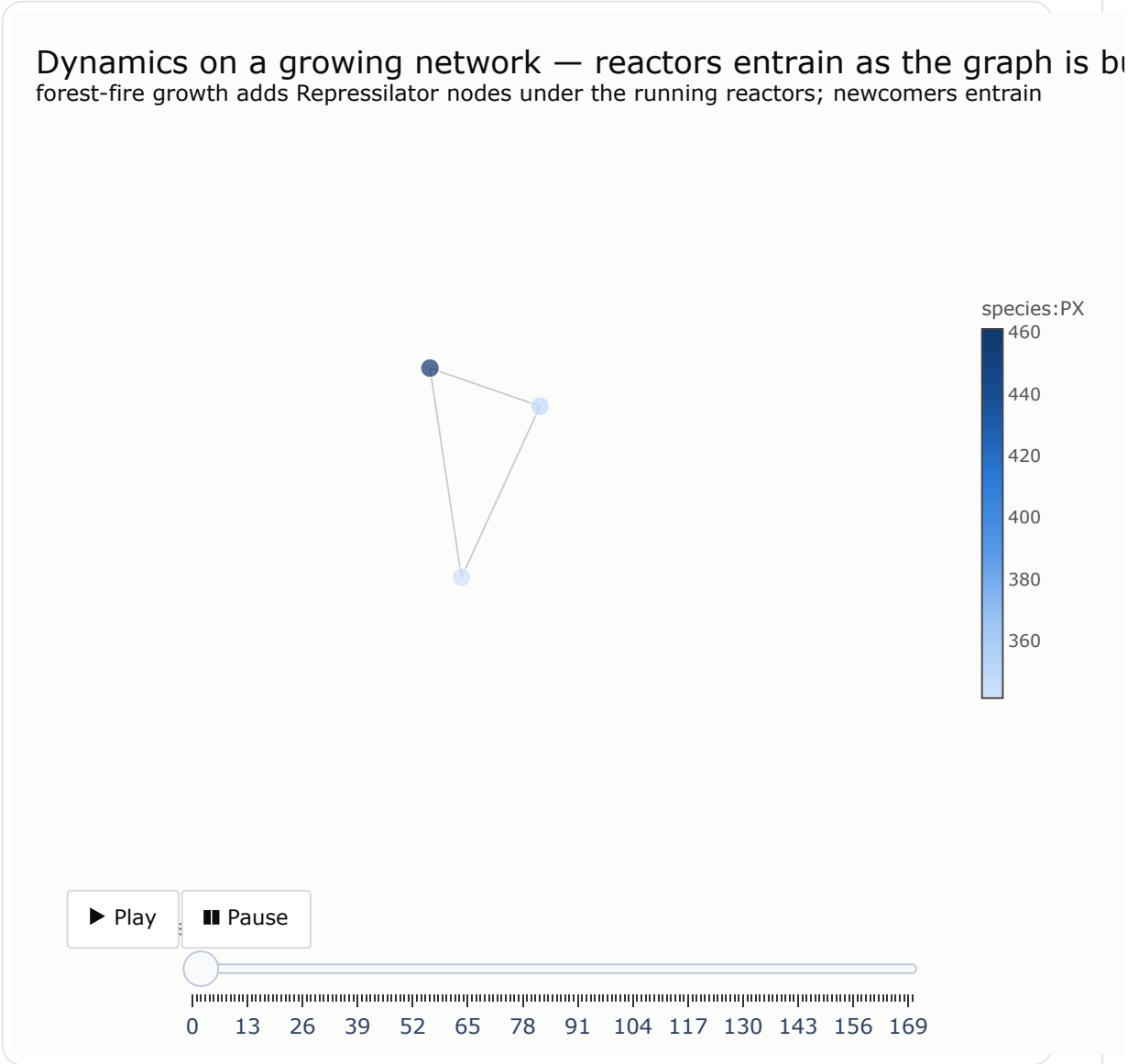
**Model & key assumptions**

**Methods · Simulations**

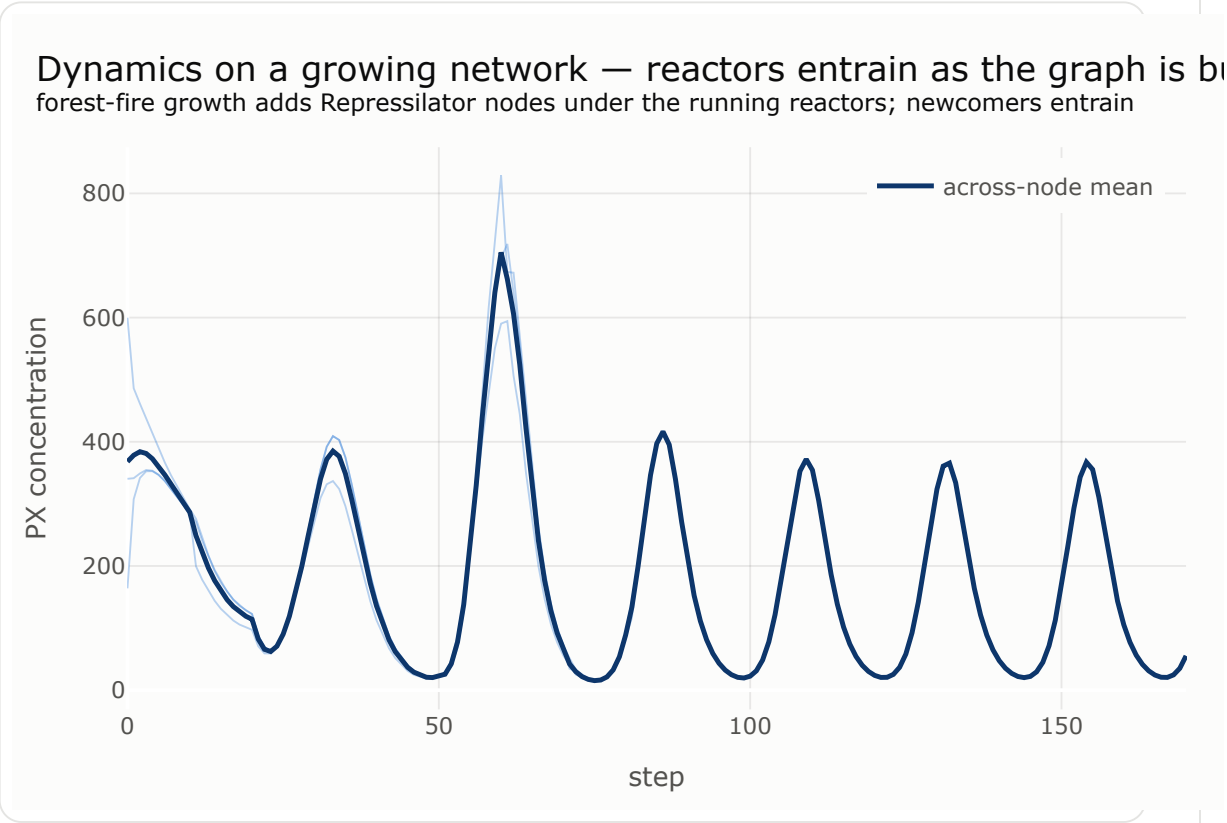
Model `growing_repressilator_network`

**Visualizations**

GROWING REPRESSILATOR NETWORK



GROWING REPRESSILATOR NETWORK TIMESERIES




**Finding**

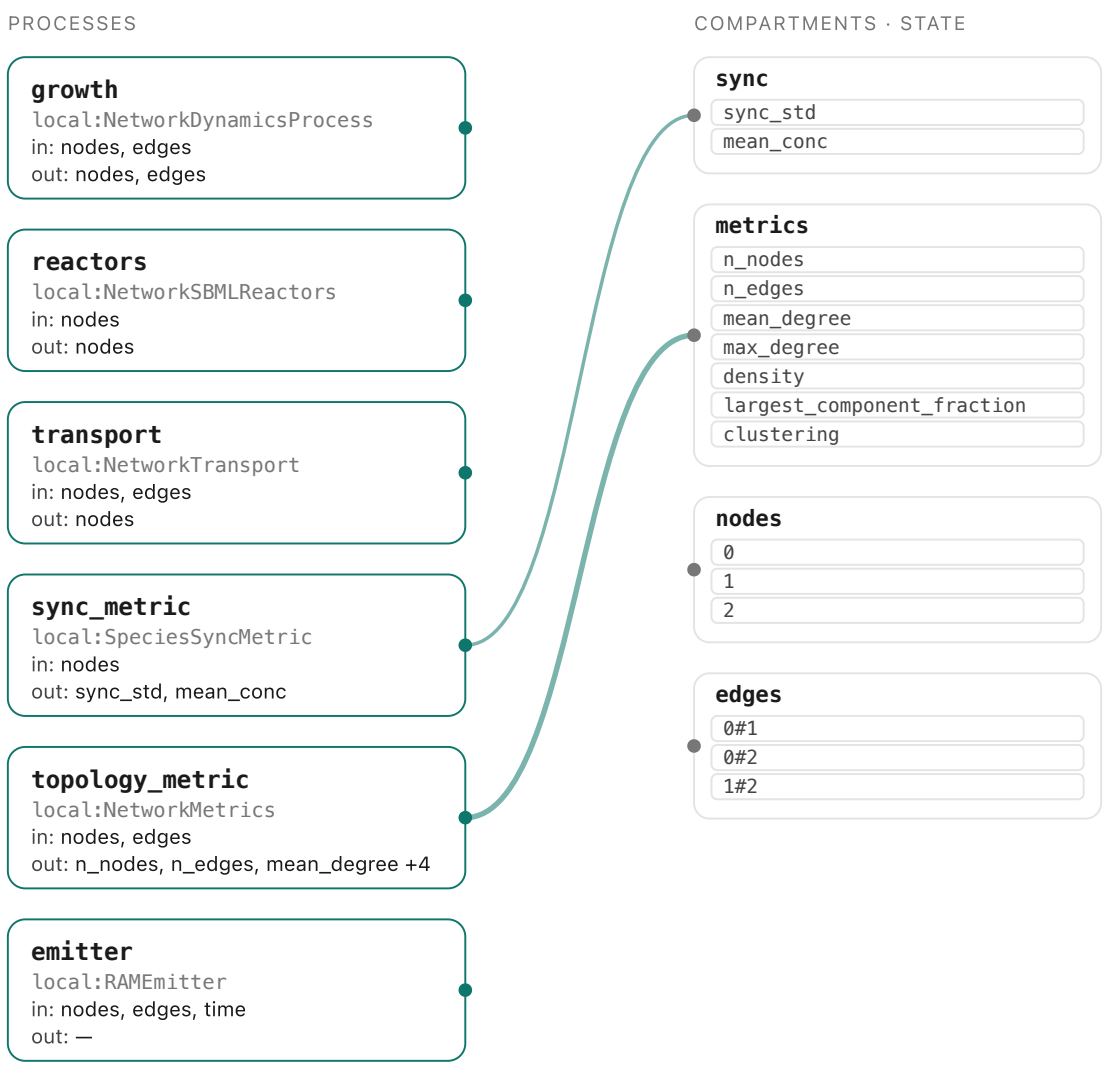
finding · Plausible

**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_c.growing_repressilator_network` — 6 processes wired to 4 compartments over 2 links. ● writes · ● reads.

`viva_dynamic_networks.composites.family_c.growing_repressilator_network`

**Simulation run**

Reproduce: `vwb run study 11-coevolving-growing-network`

**Tests**

**Across-node PX spread stays well below the oscillation amplitude even as the network grows (most nodes entrained).** `at_most 150` pending

`grows-and-mostly-entrains · quantitative · sync_std`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

## 12 Dynamics on a rewiring network — synchrony survives continual rewiring

ACCEPTED

*Do the coupled oscillators stay synchronised when the very edges that couple them are continuously rewired?*

**Hypothesis & discriminating control**

With the topology continuously small-world-rewired under the running reactors (node and edge counts conserved), the coupling remains strong enough to reach and hold synchrony.

MODEL EXPERIMENT Run the rewiring_repressilator_network composite — SBML reactors + transport on a topology that a PBG reaction reconfigures during the run — and measure synchrony.

OBSERVABLE `1 acceptance test`

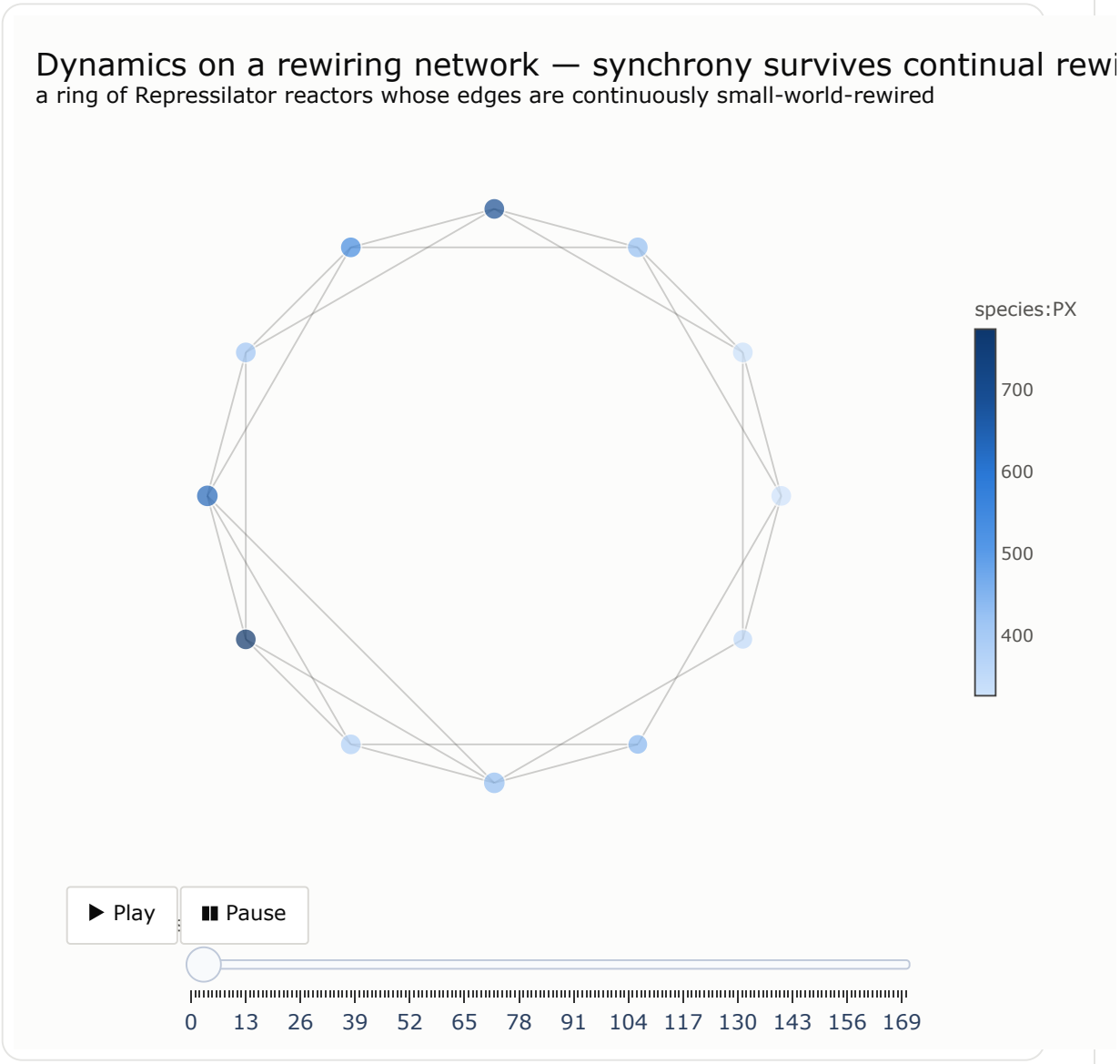
**Model & key assumptions**

**Methods · Simulations**

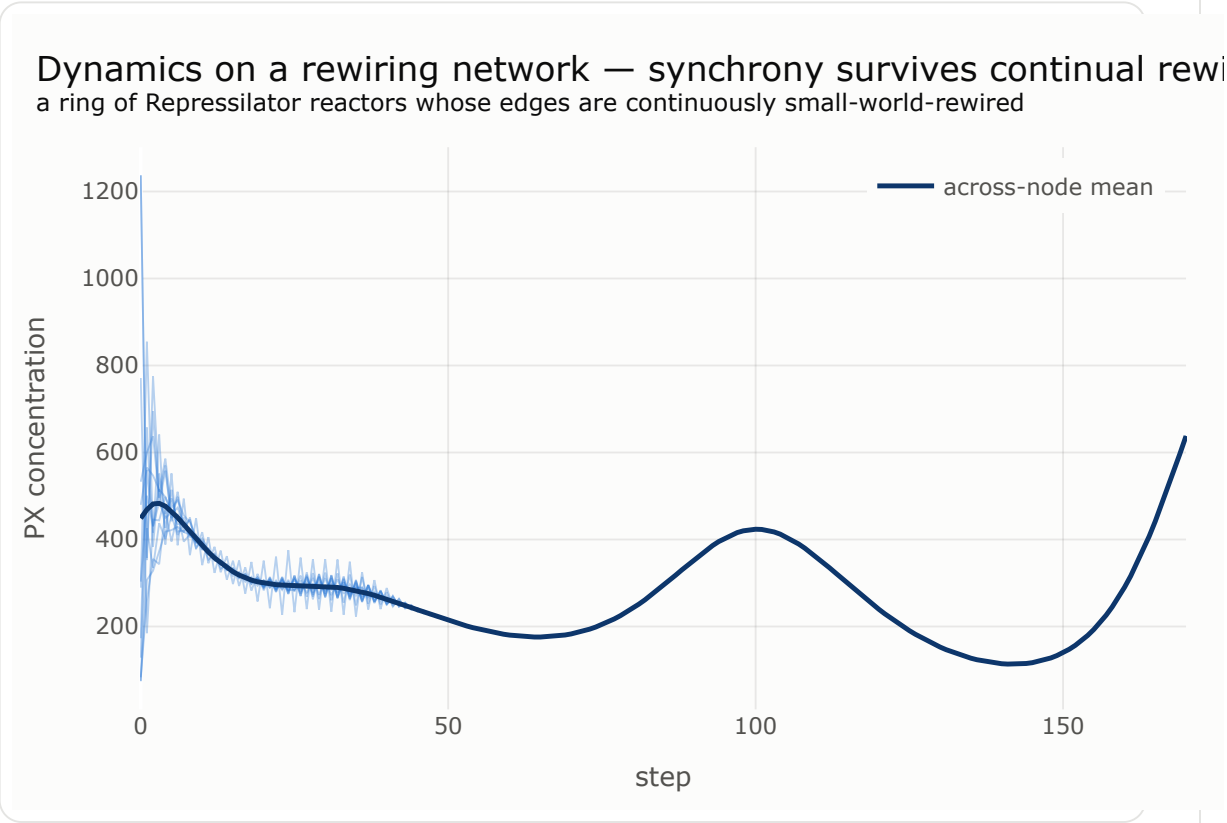
Model `rewiring_repressilator_network`

**Visualizations**

REWIRING REPRESSILATOR NETWORK



REWIRING REPRESSILATOR NETWORK TIMESERIES




**Finding**

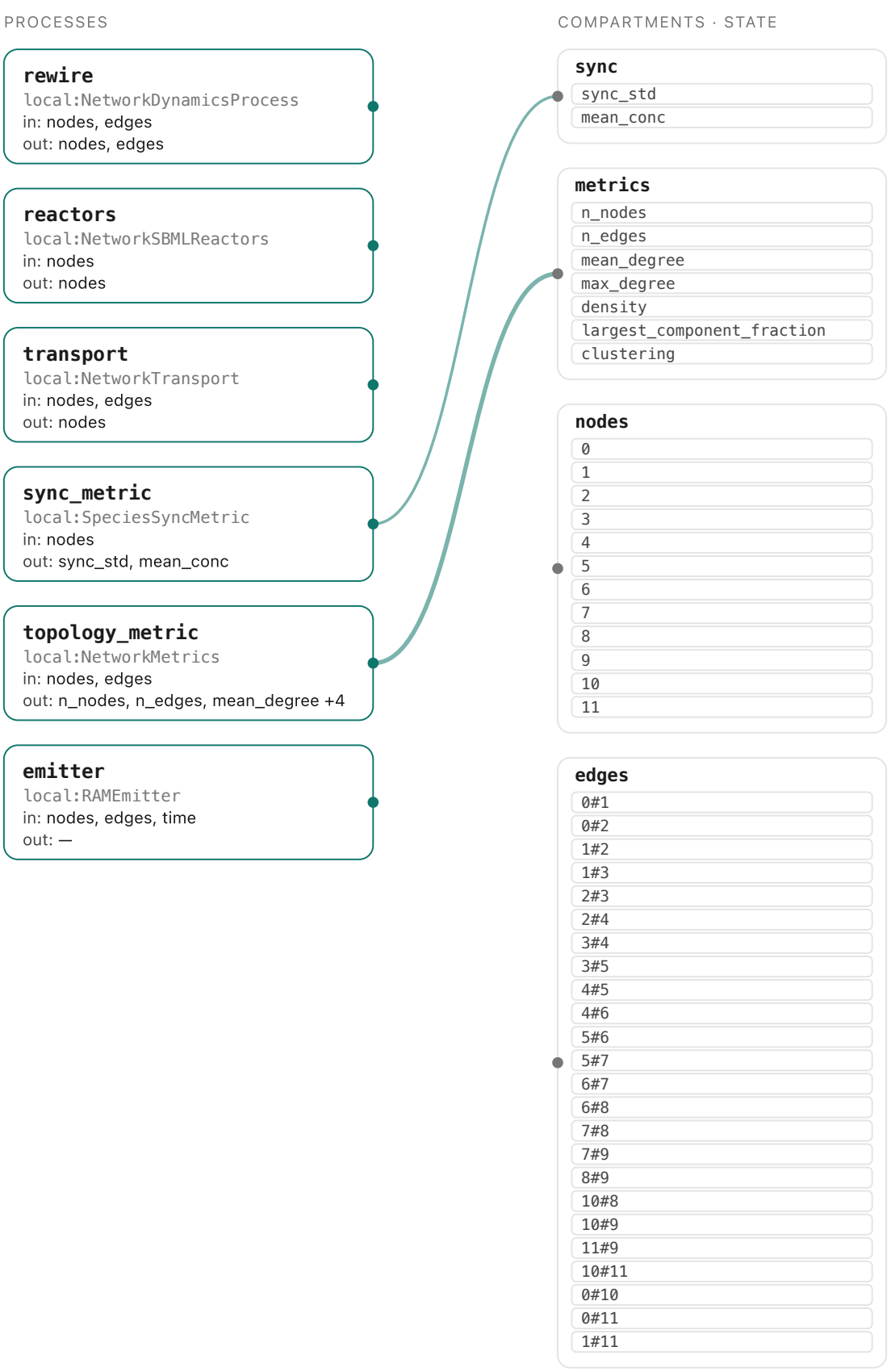
finding · Plausible

**Verdict**

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_c.rewiring_repressilator_network` — 6 processes wired to 4 compartments over 2 links. ● writes · ● reads.

`viva_dynamic_networks.composites.family_c.rewiring_repressilator_network`

**Simulation run**

Reproduce: `vwb run study 12-coevolving-rewiring-network`

**Tests**

Final across-node PX spread is <30% of the early value despite continual rewiring. `at_most 0.3` pending

`rewires-and-synchronises · quantitative · sync_std`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

## 13 Quorum-sensing coupling (proposed)

● PLANNED

*Does coupling through a single diffusible signal molecule synchronise the networked reactors as effectively as diffusing the internal proteins?*

**Hypothesis & discriminating control**

A dedicated autoinducer species produced and sensed per node, diffusing along the edges, provides indirect coupling sufficient for synchronisation at lower transport flux than direct protein diffusion.

MODEL EXPERIMENT Extend the node SBML model with a signaling species and diffuse only that molecule; compare synchronisation to study 09.

**Model & key assumptions**

**Methods · Simulations**

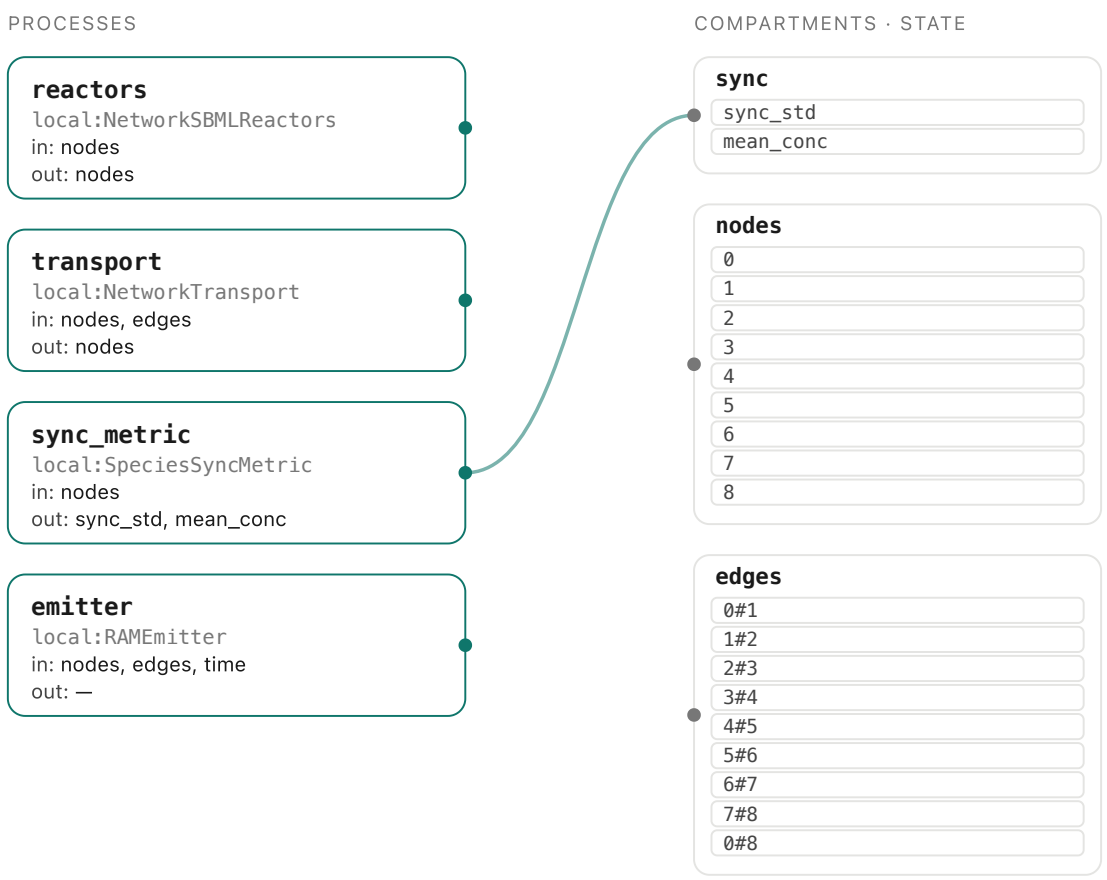
Model `repressilator_network_coupled`

**Verdict**

● PLANNED **Not yet run** · Planned

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_b.repressilator_network_coupled` — 4 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_b.repressilator_network_coupled`

**Simulation run**

Reproduce: `vwb run study 13-quorum-sensing-coupling`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

## ▾ 14 Model-agnostic node reactor — Brusselator (proposed)

● PLANNED

*Is the SBML-at-nodes architecture model-agnostic — does swapping the Repressilator for a Brusselator reproduce reaction–diffusion patterning on the network with no code change?*

**Hypothesis & discriminating control**

Because NetworkSBMLReactors loads any SBML model and NetworkTransport diffuses any species, pointing them at a Brusselator yields Turing-like patterning / synchronisation on the network with the identical process code.

MODEL EXPERIMENT Vendor a Brusselator SBML, point the reactors at it, and reproduce a Family-B/C study to demonstrate model-independence.

**Model & key assumptions**

**Methods · Simulations**

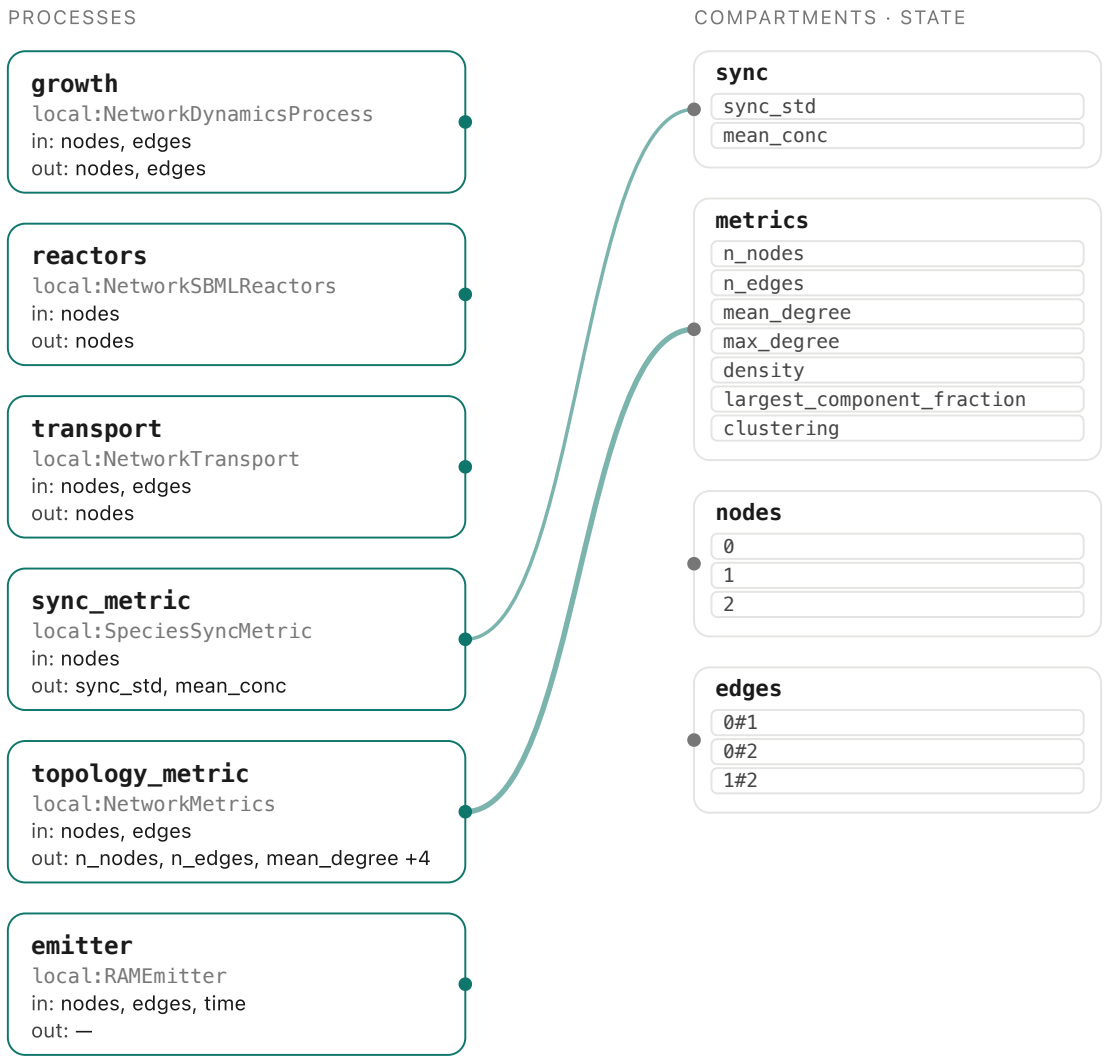
Model `growing_repressilator_network`

**Verdict**

● PLANNED **Not yet run** · Planned

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_dynamic_networks.composites.family_c.growing_repressilator_network` — 6 processes wired to 4 compartments over 2 links. ● writes · ● reads.

`viva_dynamic_networks.composites.family_c.growing_repressilator_network`

**Simulation run**

Reproduce: `vwb run study 14-brusselator-model-agnostic`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

## ▾ 15 SBML-driven adaptive coevolution (proposed)

● PLANNED

*When the continuous node state (an SBML readout) itself drives the rewiring, does a self-organised modular topology emerge?*

**Hypothesis & discriminating control**

Letting each node rewire toward neighbours with a similar SBML observable (e.g. oscillator phase) closes the state→topology feedback and self-organises the network into phase-coherent modules.

MODEL EXPERIMENT Couple an SBML-observable-driven rewiring rule to the Family-B reactors, so topology and continuous state co-adapt (the continuous generalisation of study 07).

**Model & key assumptions**

**Methods · Simulations**

Model `adaptive_rewire`

**Verdict**

● PLANNED **Not yet run** · Planned

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**


PROCESSES
network
local:NetworkDynamicsProcess
in: nodes, edges
out: nodes, edges
metrics_process
local:NetworkMetrics
in: nodes, edges
out: n_nodes, n_edges, mean_degree +4
emitter
local:RAMEmitter
in: nodes, edges, time
out: —
COMPARTMENTS · STATE
metrics
n_nodes
n_edges
mean_degree
max_degree
density
largest_component_fraction
clustering
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Static bigraph of `viva_dynamic_networks.composites.family_a.adaptive_rewire` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.adaptive_rewire`

**Simulation run**

Reproduce: `vwb run study 15-sbml-adaptive-coevolution`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

# Synthesis & gaps

Where results agree, conflict, or leave open claims — and the priority follow-ups that would resolve them.

## Contradictions & uncertainty

CAUSAL CONTINUITY

These studies take their inputs from outside the chain (a supplied feed or reset state) rather than from an upstream study's output — so their result does not yet rest on a modeled upstream mechanism:

Erdős–Rényi random growth

## Next studies & priorities

### P1 Quorum-sensing coupling (proposed)

Extend the node SBML model with a signaling species and diffuse only that molecule; compare synchronisation to study 09.

— 13-quorum-sensing-coupling

### P1 Model-agnostic node reactor — Brusselator (proposed)

Vendor a Brusselator SBML, point the reactors at it, and reproduce a Family-B/C study to demonstrate model-independence.

— 14-brusselator-model-agnostic

### P1 SBML-driven adaptive coevolution (proposed)

Couple an SBML-observable-driven rewiring rule to the Family-B reactors, so topology and continuous state co-adapt (the continuous generalisation of study 07).

— 15-sbml-adaptive-coevolution

OPEN EXPERT DECISIONS (1)

1. Build out the proposed follow-ups (studies 13–15)?

# Validation & reproducibility

Automated, code-computed consistency checks (no AI): conservation/invariant ledgers, computed-vs-authored verdicts, run reproducibility (environment + determinism), parameter drift and acceptance-criterion coverage. Each study's own ledger predicates (e.g. mass/charge balance) are its gating behavior tests, listed on the study's card (its Tests).

✓ No open validation flags — the deterministic scan found nothing to act on.

▼ REPORT READINESS — 152 FINDINGS ACROSS 14 CHECKS (27 ERRORS, 125 WARNINGS)

**12 finding without statement** error

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

**12 visualization class misplaced** error

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

**3 missing visualizations** error

`13-quorum-sensing-coupling, 14-brusselator-model-agnostic, 15-sbml-adaptive-coevolution`

**20 gate class missing** warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

**15 status legacy only** warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +11`

**14 dag edges legacy only** warning

`02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation, 05-duplication-divergence +10`

**12 missing conditions block** warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

**12 missing planned runs** warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

**12 missing readouts** warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

**12 status claims done no runs recorded** warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

**12 missing simulation set** warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

**12 missing variants** warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

**3 visualization gap** warning

`13-quorum-sensing-coupling, 14-brusselator-model-agnostic, 15-sbml-adaptive-coevolution`

**1 investigation missing adversarial** warning

`dynamic-networks`

## Claim–evidence map

Each study's central scientific claim, its evidence status, and a link to the study.

| CLAIM | Status | Study |
|---|---|---|
| Random growth produces a homogeneous, hub-free network (Poisson-like degrees) — the null baseline the structured rules are read against. | SUPPORTED | Erdős–Rényi random growth |
| Degree-proportional attachment produces scale-free hubs with a power-law degree tail — unlike random growth. | SUPPORTED | Barabási–Albert preferential attachment |
| Random rewiring of a ring lattice preserves edge count while shortcuts collapse path length — producing the small-world regime. | SUPPORTED | Watts–Strogatz small-world rewiring |
| Fitness-weighted attachment still builds hubs, but a high-fitness latecomer can overtake older nodes — unlike pure Barabási–Albert. | SUPPORTED | Bianconi–Barabási fitness attachment |
| Copy-then-diverge growth reproduces the heavy-tailed, locally clustered topology seen in protein-interaction networks. | SUPPORTED | Duplication–divergence (biological growth) |
| Forest-fire growth densifies the network: mean degree rises as the graph grows, reproducing the densification law. | SUPPORTED | Forest-fire growth & densification |
| Coevolving opinions and rewiring self-organise the network into like-minded clusters — topology and state adapt to each other. | SUPPORTED | Adaptive coevolution (opinions + rewiring) |
| Activity-driven firing produces a sparse, blinking temporal network — each snapshot is instantaneous, unlike the accreting Family-A graphs. | SUPPORTED | Activity-driven temporal network |
| With an SBML Repressilator at each node and protein diffusion along the edges, the coupled oscillators synchronise — across-node concentration spread collapses. | SUPPORTED | SBML reactors on a network — diffusive coupling synchronises |
| With diffusion off, the per-node oscillators remain independent and out of phase — the control that isolates coupling as the cause of synchronisation in study 09. | SUPPORTED | SBML reactors on a network — uncoupled control (no diffusion) |
| Continuous SBML dynamics run correctly on a network being built node-by-node by a PBG reaction: the system stays bounded and oscillating, and most newcomers entrain through the dense local coupling — but synchronisability is topology-dependent, with weakly-attached (low-degree) nodes lagging the syn | SUPPORTED | Dynamics on a growing network — reactors entrain as the graph is built |
| Coupled SBML oscillators synchronise and stay synchronised even as the coupling edges are continuously rewired — reaction–diffusion is robust to an evolving (but well-connected) topology. | SUPPORTED | Dynamics on a rewiring network — synchrony survives continual rewiring |

## Evidence register

Every study in the campaign (15) — scientific verdict, run status, and one-line result, each linked to its full study card above.

| STUDY | Verdict | Run | Result |
|---|---|---|---|
| Erdős–Rényi random growth | SUPPORTED | ran | Random growth produces a homogeneous, hub-free network (Poisson-like degrees) — the null baseline the structured rules are read against. |

| STUDY | Verdict | Run | Result |
|---|---|---|---|
| Barabási–Albert preferential attachment | SUPPORTED | ran | Degree-proportional attachment produces scale-free hubs with a power-law degree tail — unlike random growth. |
| Watts–Strogatz small-world rewiring | SUPPORTED | ran | Random rewiring of a ring lattice preserves edge count while shortcuts collapse path length — producing the small-world regime. |
| Bianconi–Barabási fitness attachment | SUPPORTED | ran | Fitness-weighted attachment still builds hubs, but a high-fitness latecomer can overtake older nodes — unlike pure Barabási–Albert. |
| Duplication–divergence (biological growth) | SUPPORTED | ran | Copy-then-diverge growth reproduces the heavy-tailed, locally clustered topology seen in protein-interaction networks. |
| Forest-fire growth & densification | SUPPORTED | ran | Forest-fire growth densifies the network: mean degree rises as the graph grows, reproducing the densification law. |
| Adaptive coevolution (opinions + rewiring) | SUPPORTED | ran | Coevolving opinions and rewiring self-organise the network into like-minded clusters — topology and state adapt to each other. |
| Activity-driven temporal network | SUPPORTED | ran | Activity-driven firing produces a sparse, blinking temporal network — each snapshot is instantaneous, unlike the accreting Family-A graphs. |
| SBML reactors on a network — diffusive coupling synchronises | SUPPORTED | ran | With an SBML Repressilator at each node and protein diffusion along the edges, the coupled oscillators synchronise — across-node concentration spread collapses. |
| SBML reactors on a network — uncoupled control (no diffusion) | SUPPORTED | ran | With diffusion off, the per-node oscillators remain independent and out of phase — the control that isolates coupling as the cause of synchronisation in study 09. |
| Dynamics on a growing network — reactors entrain as the graph is built | SUPPORTED | ran | Continuous SBML dynamics run correctly on a network being built node-by-node by a PBG reaction: the system stays bounded and oscillating, and most newcomers entrain through the dense local coupling — but synchronisabilit |
| Dynamics on a rewiring network — synchrony survives continual rewiring | SUPPORTED | ran | Coupled SBML oscillators synchronise and stay synchronised even as the coupling edges are continuously rewired — reaction–diffusion is robust to an evolving (but well-connected) topology. |
| Quorum-sensing coupling (proposed) | PENDING | planned | Extend the node SBML model with a signaling species and diffuse only that molecule; compare synchronisation to study 09. |
| Model-agnostic node reactor — Brusselator (proposed) | PENDING | planned | Vendor a Brusselator SBML, point the reactors at it, and reproduce a Family-B/C study to demonstrate model-independence. |
| SBML-driven adaptive coevolution (proposed) | PENDING | planned | Couple an SBML-observable-driven rewiring rule to the Family-B reactors, so topology and continuous state co-adapt (the continuous generalisation of study 07). |

# References

The papers behind each acceptance band, finding and literature anchor.


**Elowitz MB 2000** `elowitz2000repressilator` — A synthetic oscillatory network of transcriptional regulators · Elowitz MB, Leibler S, 2000 · source ↗ 4×

**Leskovec J 2007** `leskovec2007graph` — Graph evolution: Densification and shrinking diameters · Leskovec J, Kleinberg J, Faloutsos C, 2007 · source ↗ 3×

**Watts DJ 1998** `watts1998smallworld` — Collective dynamics of 'small-world' networks · Watts DJ, Strogatz SH, 1998 · source ↗ 3×

**Albert R 2002** `albert2002statistical` — Statistical mechanics of complex networks · Albert R, Barabási A-L, 2002 · source ↗ 2×

**Barabási A-L 1999** `barabasi1999emergence` — Emergence of scaling in random networks · Barabási A-L, Albert R, 1999 · source ↗ 2×

**Bianconi G 2001** `bianconi2001competition` — Competition and multiscaling in evolving networks · Bianconi G, Barabási A-L, 2001 · source ↗ 2×

**Colizza V 2007** `colizza2007reaction` — Reaction–diffusion processes and metapopulation models in heterogeneous networks · Colizza V, Pastor-Satorras R, Vespignani A, 2007 · source ↗ 2×

**Erdős P 1960** `erdos1960evolution` — On the evolution of random graphs · Erdős P, Rényi A, 1960 2×

**Gross T 2008** `gross2008adaptive` — Adaptive coevolutionary networks: a review · Gross T, Blasius B, 2008 · source ↗ 2×

**Ispolatov I 2005** `ispolatov2005duplication` — Duplication-divergence model of protein interaction network · Ispolatov I, Krapivsky PL, Yuryev A, 2005 · source ↗ 2×

**Perra N 2012** `perra2012activity` — Activity driven modeling of time varying networks · Perra N, Gonçalves B, Pastor-Satorras R, Vespignani A, 2012 · source ↗ 2×

**Holme P 2012** `holme2012temporal` — Temporal networks · Holme P, Saramäki J, 2012 · source ↗ 1×

**MEJ 2003** `newman2003structure` — The structure and function of complex networks · Newman MEJ, 2003 · source ↗ 1×

**Rodrigues FA 2016** `rodrigues2016kuramoto` — The Kuramoto model in complex networks · Rodrigues FA, Peron TKDM, Ji P, Kurths J, 2016 · source ↗ 1×

**Vázquez A 2003** `vazquez2003modeling` — Modeling of protein interaction networks · Vázquez A, Flammini A, Maritan A, Vespignani A, 2003 1×


# Glossary

**reaction (process-bigraph)**
A structural rewrite of the bigraph emitted as _add/_remove sentinels on a map port; here it adds/removes network nodes and edges.

**degree**
Number of edges incident to a node.

**scale-free**
A network whose degree distribution has a power-law tail — a few very-high-degree hubs.

**small-world**

High local clustering with short global path lengths.

**densification**
Edge count growing super-linearly in node count as a network evolves.

Leave an overall note, and per-study notes on each study card. Then **Review & export** assembles them into a YAML feedback report.

FEEDBACK · THE WHOLE REPORT 👍 Looks right ✏️ Needs revision Add a note…

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Rendered from workspace `viva-dynamic-networks` · deterministic from investigation + study YAML.

`investigations/dynamic-networks/investigation.yaml · studies/*/study.yaml · loop-trajectory JSON`